

	Curriculum Document			
Curriculum Code	Curriculum Title			
252901001	Occupational Certificate: Cybersecurity Analyst			
	Name	Email	Phone	Logo
Development Quality Partner	MICT SETA	Gugu.Sema@mict.org.za	011-2072600	

Learner QDF Signature

Date

QDF Signature

Date

DQP Representative Signature

Date

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2. 252901-001-00-KM-02, Fundamentals of Network Security and Defence (Network Defender), NQF Level 5, Credits 12	12
3. 252901-001-00-KM-03, Cybersecurity and Cyber Threats and Attacks (Ethical Hacking), NQF Level 5, Credits 12	15
4. 252901-001-00-KM-04, Introduction to Cybersecurity Governance, Legislation and Ethics, NQF Level 4, Credits 4	24
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SECTION 1: CURRICULUM SUMMARY

1. Occupational Information

1.1 Associated Occupation

252901: ICT Security Specialist

1.2 Occupation or Specialisation Addressed by this Curriculum

252901001: Cybersecurity Analyst

1.3 Alternative Titles used by Industry

- None

2. Curriculum Information

2.1 Curriculum Structure

This qualification is made up of the following compulsory Knowledge and Practical Skills Modules:

Knowledge Modules:

- 252901001-KM-01, Introduction to Cybersecurity, NQF Level 4, Credits 8
- 252901001-KM-02, Fundamentals of Network Security and Defence (Network Defender), NQF Level 5, Credits 12
- 252901001-KM-03, Cybersecurity and Cyber Threats and Attacks (Ethical Hacking), NQF Level 5, Credits 12
- 252901001-KM-04, Introduction to Cybersecurity Governance, Legislation and Ethics, NQF Level 4, Credits 4
- 252901001-KM-05, Fundamentals of Design Thinking and Innovation, NQF Level 4, Credits 1
- 252901001-KM-06, Refresher: Logical Thinking and Basic Calculations, NQF Level 4, Credits 3
- 252901001-KM-07, Computers, Devices and Computing Systems, NQF Level 4, Credits 6
- 252901001-KM-08, Data and Databases Vulnerabilities, NQF Level 4, Credits 3
- 252901001-KM-09, Introduction to 4IR and Future Skills, NQF Level 4, Credits 4

Total number of credits for Knowledge Modules: 53

Practical Skills Modules:

- 252901001-PM-01, Ensure Compliance in terms of Legal Cybersecurity Requirements and National and International Standards, NQF Level 5, Credits 4
- 252901001-PM-02, Assess Risks and Vulnerabilities and Evaluate Current Security Measures, NQF Level 5, Credits 20

- 252901001-PM-03, Implement Protection, Prevention and Detection Measures to Mitigate Risk, Violations and Vulnerabilities, NQF Level 5, Credits 20
- 252901001-PM-04, Apply Logical Thinking and Maths: Refresher, NQF Level 4, Credits 6
- 252901001-PM-05, Apply Basic Scriptwriting for Cybersecurity Toolsets, NQF Level 4, Credits 4
- 252901001-PM-06, Access and Visualise Structured Data Using Spreadsheets: Refresher, NQF Level 4, Credits 5
- 252901001-PM-07, Applying Design Thinking Methodologies, NQF Level 4, Credits 4
- 252901001-PM-08, Function Ethically and Effectively as a Member of a Multidisciplinary Team, NQF Level 4, Credits 5

Total number of credits for Practical Skills Modules: 68

This qualification also requires the following Work Experience Modules:

- 252901001-WM-01, Compliance with Legal Cybersecurity Requirements, NQF Level 5, Credits 12
- 252901001-WM-02, Cybersecurity Risk Assessment and Mitigation, NQF Level 5, Credits 20
- 252901001-WM-03, Cybersecurity Detection, Protection and Prevention Processes, NQF Level 5, Credits 20

Total number of credits for Work Experience Modules: 52

2.2 Entry Requirements

NQF 4

3. Assessment Quality Partner Information

Name of body: MICT Seta

Address of body: 19 Richards Dr, Halfway House, Midrand, 1685

Contact person name: Gugu Sema

Contact person work telephone number: 011-2072600

4. Part Qualification Curriculum Structure

None

SECTION 2: OCCUPATIONAL PROFILE

1. Occupational Purpose

Cybersecurity Analysts apply the practice of protecting assets such as networks, computer systems and information assets from malicious attacks and threats. They assess and mitigate risks and potential intrusions and identify risks and vulnerabilities. They study existing techniques for managing security issues and maintaining the security of information and systems in the working environment ensuring legal compliance

2. Occupational Tasks

- Demonstrate knowledge and understanding of cybersecurity concepts and investigate how cybersecurity affects legal compliance and solidarity in companies and communities (NQF Level 5)
- Assess risk to assets and evaluate current cybersecurity protection measures (NQF Level 5)
- Implement detection, protection and prevention systems and respond to breaches or incidences (NQF Level 5)

3. Occupational Task Details

3.1. Demonstrate knowledge and understanding of cybersecurity concepts and investigate how cybersecurity affects legal compliance and solidarity in companies and communities (NQF Level 5)

Unique Product or Service:

- Legal compliance in cybersecurity

Occupational Responsibilities:

- Ensure Compliance in terms of Legal Cybersecurity Requirements and National and International Standards

Occupational Contexts:

- Compliance with Legal Cybersecurity Requirements

3.2. Assess risk to assets and evaluate current cybersecurity protection measures (NQF Level 5)

Unique Product or Service:

- Risk, vulnerabilities and threats identified

Occupational Responsibilities:

- Assess Risks and Vulnerabilities and Current Security Measures

Occupational Contexts:

- Cybersecurity Risk Assessment and Mitigation

3.3. Implement detection, protection and prevention systems and respond to breaches or incidences (NQF Level 5)

Unique Product or Service:

- Protection of company assets

Occupational Responsibilities:

- Implement Protection, Prevention and Detection Measures to Mitigate Risk, Violations and Vulnerabilities

Occupational Contexts:

- Cybersecurity Detection, Protection and Prevention Processes

SECTION 3: CURRICULUM COMPONENT SPECIFICATIONS

SECTION 3A: KNOWLEDGE MODULE SPECIFICATIONS

List of Knowledge Modules for which Specifications are included

Knowledge	252901-001-00-KM-01	Introduction to Cybersecurity	4	8
Knowledge	252901-001-00-KM-02	Fundamentals of Network Security and Defence (Network Defender)	5	12
Knowledge	252901-001-00-KM-03	Cybersecurity and Cyber Threats and Attacks (Ethical Hacking)	5	12
Knowledge	252901-001-00-KM-04	Introduction to Cybersecurity Governance, Legislation and Ethics	4	4
Knowledge	252901-001-00-KM-05	Fundamentals of Design Thinking and Innovation	4	1
Knowledge	252901-001-00-KM-06	Refresher: Logical Thinking and Basic Calculations	4	3
Knowledge	252901-001-00-KM-07	Computers, Devices and Computing Systems	4	6
Knowledge	252901-001-00-KM-08	Data and Databases Vulnerabilities	4	3
Knowledge	252901-001-00-KM-09	Introduction to 4IR and Future Skills	4	4

1. 252901-001-00-KM-01, Introduction to Cybersecurity, NQF Level 4, Credits 8

1.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of fundamentals of various computer and network security threats such as identity theft, credit card fraud, phishing, virus and backdoors, emails hoaxes, loss of confidential information, hacking attacks, and social engineering

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|--|------------|
| • KM-01-KT01 | : Introduction to computer and mobile device security | 15% |
| • KM-01-KT02 | : Various computer and network security threats | 20% |
| • KM-01-KT03 | : Identity theft | 10% |
| • KM-01-KT04 | : Adopting good cybersecurity practices | 15% |
| • KM-01-KT05 | : Safeguard mobile, media and social networking profiles as user | 10% |
| • KM-01-KT06 | : Protecting computers, accounts and data as user | 20% |
| • KM-01-KT07 | : Understand security incidents and reporting | 10% |

1.2 Guidelines for Topics

1.2.1 KM-01-KT01 : Introduction to computer and mobile device security **15%**

Topic elements to be covered include:

- KT0101 Governance and legislation
- KT0102 Security policy
- KT0103 Physical security (e.g. biometric authentication)
- KT0104 Web content filters
- KT0105 Need for protection of privacy and data

Internal Assessment Criteria and Weight

- IAC0101 The importance of computer and mobile device security is justified.
- IAC0102 Objectives of computer and mobile device security are explained.

(Weight 15%)

1.2.2 KM-01-KT02 : Various computer and network security threats **20%**

Topic elements to be covered include:

- KT0201 Malware.
- KT0202 Viruses.
- KT0203 Spyware.
- KT0204 Adware.
- KT0205 Trojan horses.
- KT0206 Worms.
- KT0207 Phishing.
- KT0208 Spear phishing.

- KT0209 Insider security threats

Internal Assessment Criteria and Weight

- IAC0201 Computer and network security threats and attacks are defined and how they work is explained.
- IAC0202 Potential solutions are identified.

(Weight 20%)

1.2.3 KM-01-KT03 : Identity theft 10%
Topic elements to be covered include:

- KT0301 Phishing scams,
- KT0302 Malware,
- KT0303 Social engineering,
- KT0304 Financial frauds

Internal Assessment Criteria and Weight

- IAC0301 An understanding of types of identity theft and how to mitigate them is demonstrated

(Weight 10%)

1.2.4 KM-01-KT04 : Adopting good cybersecurity practices 15%
Topic elements to be covered include:

- KT0401 Use of registered software
- KT0402 Update software when they become old to newer versions which tend to have more security upgrades
- KT0403 Encrypt data
- KT0404 Use strong authentication (passwords and credentials) for all accounts
- KT0405 Change passwords often and make sure they are strong and not easy to guess
- KT0406 Data backup.

Internal Assessment Criteria and Weight

- IAC0401 The importance of adhering to good cybersecurity practices is justified.

(Weight 15%)

1.2.5 KM-01-KT05 : Safeguard mobile, media and social networking profiles as user 10%
Topic elements to be covered include:

- KT0501 Security on social networking sites
- KT0502 Securing mobile devices

Internal Assessment Criteria and Weight

- IAC0501 The importance of safeguarding mobile and media devices is justified.
- IAC0502 Mechanisms to safeguard mobile and media devices are discussed.

(Weight 10%)

1.2.6 KM-01-KT06 : Protecting computers, accounts and data as user 20%
Topic elements to be covered include:

- KT0601 Securing operating systems
- KT0602 Internet security
- KT0603 Securing email communications
- KT0604 Securing the cloud
- KT0605 Securing network connections
- KT0606 Firewalls
- KT0607 Antivirus programs
- KT0608 Antispyware

Internal Assessment Criteria and Weight

- IAC0601 Measures of protecting computers, accounts and data are identified.
- IAC0602 The advantages of protecting computers, accounts and data are elaborated.

(Weight 20%)

1.2.7 KM-01-KT07 : Understand security incidents and reporting 10%
Topic elements to be covered include:

- KT0701 Data backup
- KT0702 disaster recovery

Internal Assessment Criteria and Weight

- IAC0701 The effect of security incidents is explained

(Weight 10%)

1.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)

- Qualification of lecturer (SME):
 - NQF 6 in industry recognised qualifications with 1 years' experience in the IT industry
 - CYBERSECURITY vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

1.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

2. 252901-001-00-KM-02, Fundamentals of Network Security and Defence NQF Level 5, Credits 12

2.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of the principles and techniques applied in the editing and proofreading processes Network Security and Defence

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|---|------------|
| • KM-02-KT01 | : Introduction to network security | 20% |
| • KM-02-KT02 | : Network risk and vulnerability management | 10% |
| • KM-02-KT03 | : Network defense fundamentals | 40% |
| • KM-02-KT04 | : Monitoring for breaches and attacks | 10% |
| • KM-02-KT05 | : Network incident response and management | 20% |

2.2 Guidelines for Topics

2.2.1 KM-02-KT01 : Introduction to network security 20%

Topic elements to be covered include:

- KT0101 Governance and legislation
- KT0102 Security policy
- KT0103 Physical security (e.g. biometric authentication)
- KT0104 Web content filters
- KT0105 Need for protection of privacy and data

Internal Assessment Criteria and Weight

IAC0101 The implications of governance and legislation on cybersecurity are reasoned.

IAC0102 The importance of protection of privacy and data is justified.

(Weight 20%)

2.2.2 KM-02-KT02 : Network risk and vulnerability management 10%

Topic elements to be covered include:

- KT0201 Network security threats, vulnerabilities, and attacks
- KT0202 Knowledge on how to protect, detect, and respond to network attacks

Internal Assessment Criteria and Weight

- IAC0201 Network risk and vulnerability management is evaluated

(Weight 10%)

2.2.3 KM-02-KT03 : Network defense fundamentals 40%

Topic elements to be covered include:

- KT0301 Network security controls, protocols, and devices
- KT0302 Physical security
- KT0303 Host security

- KT0304 Secure firewall configuration and management
- KT0305 Secure IDS configuration and management
- KT0306 Secure VPN configuration and management
- KT0307 Wireless network defense
- KT0308 Data backup and recovery

Internal Assessment Criteria and Weight

- IAC0301 Application of network security controls, protocols, perimeter appliances, secure IDS, VPN, and firewall configuration

(Weight 40%)

2.2.4 KM-02-KT04 : Monitoring for breaches and attacks

10%

Topic elements to be covered include:

- KT0401 Network traffic monitoring and analysis
- KT0402 Intricacies of network traffic signature, analysis, and vulnerability scanning

Internal Assessment Criteria and Weight

- IAC0401 Procedures for monitoring of breaches and attacks are compared

(Weight 10%)

2.2.5 KM-02-KT05 : Network incident response and management

20%

Topic elements to be covered include:

- KT0501 Incident response plan
- KT0502 Incident response process
- KT0503 Reporting and documentation
- KT0504 Lessons learnt

Internal Assessment Criteria and Weight

- IAC0501
- IAC0502

(Weight 20%)

2.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.

- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 in industry recognised qualifications with 1 years' experience in the IT industry
 - CYBERSECURITY vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

2.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

3. 252901-001-00-KM-03, Cybersecurity and Cyber Threats and Attacks NQF Level 5, Credits 12

3.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of principles of cybersecurity and Ethical Hacking and the types of threats and attacks and the respective risk.

The learning will enable learners to demonstrate an understanding of:

• KM-03-KT01	: Information security Governance and compliance	5%
• KM-03-KT02	: Information security	5%
• KM-03-KT03	: Footprinting and Reconnaissance	5%
• KM-03-KT04	: Scanning Networks	5%
• KM-03-KT05	: Enumeration	5%
• KM-03-KT06	: Vulnerability Analysis	5%
• KM-03-KT07	: System Hacking	4%
• KM-03-KT08	: Malware Threats	4%
• KM-03-KT09	: Sniffing	5%
• KM-03-KT10	: Social Engineering	4%
• KM-03-KT11	: Denial-of-Service	4%
• KM-03-KT12	: Session Hijacking	5%
• KM-03-KT13	: Evading IDS, Firewalls, and Honeypots	5%
• KM-03-KT14	: Hacking Web Servers	5%
• KM-03-KT15	: SQL Injection	5%
• KM-03-KT16	: Hacking Wireless Networks	5%
• KM-03-KT17	: Hacking Mobile Platforms	5%
• KM-03-KT18	: IoT Hacking	5%
• KM-03-KT19	: Cloud Computing	4%
• KM-03-KT20	: Cryptography	5%
• KM-03-KT21	: Cyber incident response and management	5%

3.2 Guidelines for Topics

3.2.1 KM-03-KT01 : Information security Governance and compliance 5%

Topic elements to be covered include:

- KT0101 Governance and legislation
- KT0102 Need for data protection
- KT0103 Computer network and defense fundamentals
- KT0104 Network security policy design and implementation
- KT0105 Information security laws and standards

Internal Assessment Criteria and Weight

- IAC0101 The importance of compliance with governance framework is justified.

(Weight 5%)

3.2.2 KM-03-KT02 : Information security

5%

Topic elements to be covered include:

- KT0201 Information security overview
- KT0202 Information security threats and attack vectors
- KT0203 Penetration testing concepts
- KT0204 Hacking concepts
- KT0205 Ethical hacking concepts
- KT0206 Information security controls

Internal Assessment Criteria and Weight

- IAC0201 Concepts of information security are defined
- IAC0202 Information security controls are listed and discussed.

(Weight 5%)

3.2.3 KM-03-KT03 : Footprinting and Reconnaissance

5%

Topic elements to be covered include:

- KT0301 Footprinting concepts and objectives
- KT0302 Footprinting uses
- KT0303 Types of footprinting
- KT0304 Passive and active footprinting
- KT0305 Footprinting methodologies
- KT0306 Tools, tricks and techniques for information gathering
- KT0307 Footprinting steps
- KT0308 Countermeasures and prevention
- KT0309 Footprinting Pen Testing

Internal Assessment Criteria and Weight

- IAC0301 Risks and mitigation related to footprinting and reconnaissance are interrogated.

(Weight 5%)

3.2.4 KM-03-KT04 : Scanning Networks

5%

Topic elements to be covered include:

- KT0401 Network scanning concepts and objectives
- KT0402 Types of scanning
- KT0403 Scanning methodologies, tools and techniques
- KT0404 Draw network diagrams
- KT0405 Scanning pen testing
- KT0406 Vulnerability scanning
- KT0407 Countermeasures against scanning

Internal Assessment Criteria and Weight

- IAC0401 Risks and mitigation of scanning networks are interrogated.

(Weight 5%)

3.2.5 KM-03-KT05 : Enumeration

5%

Topic elements to be covered include:

- KT0501 Enumeration concepts and objectives
- KT0502 Types of enumeration
- KT0503 Enumeration countermeasures
- KT0504 Enumeration methodologies, techniques and tools
- KT0505 Enumeration pen testing

Internal Assessment Criteria and Weight

- IAC0501 Risks and mitigation of Enumeration are interrogated.

(Weight 5%)

3.2.6 KM-03-KT06 : Vulnerability Analysis

5%

Topic elements to be covered include:

- KT0601 Vulnerability analysis concepts and objectives
- KT0602 Vulnerability assessment solutions
- KT0603 Vulnerability scoring systems
- KT0604 Vulnerability assessment tools
- KT0605 Vulnerability assessment reports

Internal Assessment Criteria and Weight

- IAC0601 Risks and mitigation of vulnerability analysis concepts and objectives are interrogated

(Weight 5%)

3.2.7 KM-03-KT07 : System Hacking

4%

Topic elements to be covered include:

- KT0701 System hacking concepts
- KT0702 Passwords, privileges, applications
- KT0703 Covering tracks
- KT0704 Penetration testing
- KT0705 Human precautions against hacking

Internal Assessment Criteria and Weight

- IAC0701 Risks and mitigation of system hacking are interrogated

(Weight 4%)

3.2.8 KM-03-KT08 : Malware Threats

4%

Topic elements to be covered include:

- KT0801 Malware concepts and objectives
- KT0802 Malware types

- KT0803 Malware analysis
- KT0804 Countermeasures and anti-malware software
- KT0805 Malware penetration testing

Internal Assessment Criteria and Weight

- IAC0801 Risks and mitigation of malware are interrogated

(Weight 4%)

3.2.9 KM-03-KT09 : Sniffing

5%

Topic elements to be covered include:

- KT0901 Sniffing concepts and objectives
- KT0902 Sniffing attacks
- KT0903 Sniffing techniques
- KT0904 Sniffing tools
- KT0905 Countermeasures
- KT0906 Detection techniques
- KT0907 Sniffing pen testing

Internal Assessment Criteria and Weight

- IAC0901 Risks and mitigation of sniffing are interrogated

(Weight 5%)

3.2.10 KM-03-KT10 : Social Engineering

4%

Topic elements to be covered include:

- KT1001 Social engineering concepts and objectives
- KT1002 Social engineering techniques
- KT1003 Types of social engineering attacks
- KT1004 Countermeasures
- KT1005 Social engineering penetration testing

Internal Assessment Criteria and Weight

- IAC1001 Risks and mitigation of social engineering are interrogated

(Weight 4%)

3.2.11 KM-03-KT11 : Denial-of-Service

4%

Topic elements to be covered include:

- KT1101 DoS/DDoS concepts and objectives
- KT1102 DoS/DDoS attack techniques and tools
- KT1103 Countermeasures and protection tools
- KT1104 DoS/DDoS attack penetration testing

Internal Assessment Criteria and Weight

- IAC1101 Risks and mitigation of DoS/DDoS are interrogated

(Weight 4%)

3.2.12 KM-03-KT12 : Session Hijacking

5%

Topic elements to be covered include:

- KT1201 Session hijacking concepts
- KT1202 Types of level session hijacking
- KT1203 Session hijacking tools
- KT1204 Countermeasures
- KT1205 Penetration testing

Internal Assessment Criteria and Weight

- IAC1201 Risks and mitigation of Session hijacking are interrogated

(Weight 5%)

3.2.13 KM-03-KT13 : Evading IDS, Firewalls, and Honeypots

5%

Topic elements to be covered include:

- KT1301 IDS, Firewall, and Honeypot Concepts
- KT1302 IDS, Firewall, and Honeypot Solutions
- KT1303 Evading IDS and Firewalls
- KT1304 IDS/Firewall Evading Tools
- KT1305 Detecting Honeypots
- KT1306 IDS/Firewall Evasion Countermeasures
- KT1307 Penetration Testing

Internal Assessment Criteria and Weight

- IAC1301 Risks and mitigation of IDS, Firewall, and Honeypot are interrogated

(Weight 5%)

3.2.14 KM-03-KT14 : Hacking Web Servers

5%

Topic elements to be covered include:

- KT1401 Web server concepts
- KT1402 Web server attacks
- KT1403 Attack tools
- KT1404 Countermeasures
- KT1405 Defense attack mechanisms
- KT1406 Security tools
- KT1407 Penetration testing and pen testing tools

Internal Assessment Criteria and Weight

- IAC1401 Risks and mitigation of hacking web servers are interrogated

(Weight 5%)

3.2.15 KM-03-KT15 : SQL Injection

5%

Topic elements to be covered include:

- KT1501 SQL injection concepts
- KT1502 Types of SQL injection
- KT1503 SQL injection methodology and tools
- KT1504 Evasion techniques
- KT1505 Countermeasures

Internal Assessment Criteria and Weight

- IAC1501 Risks and mitigation of SQL injection are interrogated

(Weight 5%)

3.2.16 KM-03-KT16 : Hacking Wireless Networks

5%

Topic elements to be covered include:

- KT1601 Wireless concepts
- KT1602 Wireless encryption
- KT1603 Wireless threats
- KT1604 Wireless hacking methodology and tools
- KT1605 Bluetooth hacking
- KT1606 Countermeasures and security tools
- KT1607 Wi-Fi Pen Testing

Internal Assessment Criteria and Weight

- IAC1601 Risks and mitigation of hacking wireless networks are interrogated

(Weight 5%)

3.2.17 KM-03-KT17 : Hacking Mobile Platforms

5%

Topic elements to be covered include:

- KT1701 Mobile platform attack vectors
- KT1702 Hacking android OS
- KT1703 Hacking iOS
- KT1704 Mobile spyware
- KT1705 Mobile device management
- KT1706 Mobile security guidelines and tools
- KT1707 Mobile pen testing

Internal Assessment Criteria and Weight

- IAC1701 Risks and mitigation of hacking mobile platforms are interrogated

(Weight 5%)

3.2.18 KM-03-KT18 : IoT Hacking

5%

Topic elements to be covered include:

- KT1801 IoT concepts
- KT1802 IoT attacks
- KT1803 IoT hacking methodology and tools
- KT1804 Countermeasures
- KT1805 IoT pen testing

Internal Assessment Criteria and Weight

- IAC1801 Risks and mitigation of IoT hacking are interrogated

(Weight 5%)

3.2.19 KM-03-KT19 : Cloud Computing

4%

Topic elements to be covered include:

- KT1901 Cloud computing concepts
- KT1902 Cloud computing threats and attacks
- KT1903 Cloud security
- KT1904 Cloud security tools
- KT1905 Cloud penetration testing

Internal Assessment Criteria and Weight

- IAC1901 Risks, threats and vulnerabilities and mitigation of cloud computing are interrogated

(Weight 30%)

3.2.20 KM-03-KT20 : Cryptography

5%

Topic elements to be covered include:

- KT2001 Cryptography concepts and objectives
- KT2002 Types of cryptography
- KT2003 Encryption algorithms
- KT2004 Cryptography tools
- KT2005 Types of encryption
- KT2006 Cryptanalysis
- KT2007 Countermeasures

Internal Assessment Criteria and Weight

- IAC2001 Risks and mitigation of cryptography are interrogated

(Weight 5%)

3.2.21 KM-03-KT21 : Cyber incident response and management

5%

Topic elements to be covered include:

- KT2101 Incident response plan

- KT2102 Incident response process
- KT2103 Incidence response phases
 - Preparation
 - Detection and analysis
 - Containment and quarantine
 - Eradication
 - Recovery (return to production, data)
- KT2104 Post-incident activity (Lessons Learned)
 - Identify changes to security
 - Employee training
 - Weaknesses in the security system
 - Updates

Internal Assessment Criteria and Weight

- IAC2101 Concepts and principles of Cyber incident response and management are explained
- IAC2102

(Weight 5%)

3.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 in industry recognised qualifications with 1 years' experience in the IT industry
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

3.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

4. 252901-001-00-KM-04, Introduction to Cybersecurity Governance, Legislation and Ethics, NQF Level 4, Credits 4

4.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to acquire general knowledge and understanding of the various legislations governing the workplace and their implication for the employer and employees. The learning of this module will also enable the learner to acquire an understanding of the principles of areas of performance management, business planning concepts, costing of products and concepts of general ethical behaviour and its impact in the workplace

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|---------------------------------------|------------|
| • KM-04-KT01 | : Governance | 20% |
| • KM-04-KT02 | : Legislation governing workplaces | 15% |
| • KM-04-KT03 | : Introduction to ethics and security | 5% |
| • KM-04-KT04 | : Ethics at work | 14% |
| • KM-04-KT05 | : Security | 15% |
| • KM-04-KT06 | : Performance management | 10% |
| • KM-04-KT07 | : Business planning | 7% |
| • KM-04-KT08 | : Costing of products | 7% |
| • KM-04-KT09 | : Resources | 7% |

4.2 Guidelines for Topics

4.2.1 KM-04-KT01 : Governance 20%

Topic elements to be covered include:

- KT0101 Definitions and the role of governance
- KT0102 Rules, norms and actions are structured, sustained, regulated and held accountable.
- KT0103 Structures and processes and industry codes of practices
- KT0104 Transparent, participatory, inclusive and responsive
- KT0105 Domains of IT governance
- KT0106 Compliance vs non-compliance

Internal Assessment Criteria and Weight

- IAC0101 Governance as the way rules, norms and actions are structured, sustained, regulated and held accountable is justified.

(Weight 20%)

4.2.2 KM-04-KT02 : Legislation governing workplaces 15%

Topic elements to be covered include:

- KT0201 LRA
- KT0202 POPI
- KT0203 B-BBEE
- KT0204 BCEA

- KT0205 SDA
- KT0206 Current trends

Internal Assessment Criteria and Weight

- IAC0201 Legislation governing workplaces are identified and the implications for the workplace are reasoned

(Weight 15%)

4.2.3 KM-04-KT03 : Introduction to ethics and security

5%

Topic elements to be covered include:

- KT0301 Principles and practices
- KT0302 Concepts, definitions and terminology

Internal Assessment Criteria and Weight

- IAC0301 Definition, principles and practices of ethics and its influence on security within a business ecosystem, is understood

(Weight 5%)

4.2.4 KM-04-KT04 : Ethics at work

14%

Topic elements to be covered include:

- KT0401 Code of conduct and mMoral compass
- KT0402 Components of ethical behaviour, including integrity, honesty, fair dealing, respecting diversity
- KT0403 Unwritten but expected behaviours, including reliability, accountability, time keeping, respect for others
- KT0404 Lapses in ethical behaviour, including sexual harassment, racism, bullying, theft, abuse of company property, rules, time and sick leave
- KT0405 Conflicts of interest, including primary and secondary interests, the impact on individuals and organisations, and the link to corruption
- KT0406 The need for ethical behaviour and the impact or consequences of lapses in ethical behaviour
- KT0407 Copyright and plagiarism
- KT0408 Intellectual property
- KT0409 Spamming
- KT0410 Contract management
- KT0411 Software licensing
- KT0412 Pricing

Internal Assessment Criteria and Weight

- IAC0401 Concepts, issues and examples of ethical and unethical conduct are defined and described
- IAC0402 The impact of these factors on an employer and an employee is discussed

- IAC0403 The impact of lapses in ethical behaviour on the organisation and individuals in the organisation is described
- IAC0404 Processes which employer organisations use to support ethical conduct in the workplace are evaluated
- IAC0405 The importance of the responsibility to consistently comply to the industry ethics and codes of conduct within the business ecosystem, is understood and described
- IAC0406 Own moral compass is developed and applied to ensure privacy, accountability and confidentiality within the industry

(Weight 14%)

4.2.5 KM-04-KT05 : Security

15%

Topic elements to be covered include:

- KT0501 Risks, threats and vulnerabilities
- KT0502 Mitigation tools and strategies
- KT0503 Digital forensics
- KT0504 Cloud
- KT0505 Commercial law
- KT0506 Cyber security
- KT0507 New trends

Internal Assessment Criteria and Weight

- IAC0501 A sound knowledge of the various threats and vulnerabilities within the business ecosystem and the ways to mitigate these, are demonstrated
- IAC0502 The importance of cyber forensics and cybersecurity and their application within the boundaries of the law, is understood

(Weight 15%)

4.2.6 KM-04-KT06 : Performance management

10%

Topic elements to be covered include:

- KT0601 Planning, organising and control
- KT0602 Work flow
- KT0603 Cost, waste
- KT0604 Productivity, efficiency
- KT0605 Housekeeping
- KT0606 Risk, health, safety, environment and related systems
- KT0607 Quality and quality systems

- KT0608 Continual improvement

Internal Assessment Criteria and Weight

- IAC0601 Performance management as a process of ensuring that sets of activities and outputs meet an organization's goals in an effective and efficient manner, is reasoned
- IAC0602 The various focus areas of performance management (e.g., the performance of an organization, a department, an employee, or the processes) which should be in place to manage particular tasks, are identified and justified

(Weight 10%)

4.2.7 KM-04-KT07 : Business planning

7%

Topic elements to be covered include:

- KT0701 Business sustainability
- KT0702 Concept of supply and demand
- KT0703 Concept of profit, loss and breakeven
- KT0704 Accountability and responsibility
- KT0705 Competition
- KT0706 Customers
- KT0707 Contracts
- KT0708 Budgets

Internal Assessment Criteria and Weight

- IAC0701 The various concepts of business planning, including competition and budgets, are described

(Weight 7%)

4.2.8 KM-04-KT08 : Costing of products

7%

Topic elements to be covered include:

- KT0801 Input cost
- KT0802 Overhead costs
- KT0803 Direct labour cost
- KT0804 Pricing a product (Under- or over-pricing)

Internal Assessment Criteria and Weight

- IAC0801 Costs related to the production of a product are identified and the effect thereof on the price of the product is analysed

(Weight 7%)

4.2.9 KM-04-KT09 : Resources

7%

Topic elements to be covered include:

- KT0901 Human resources
- KT0902 Financial resources
- KT0903 Physical resources (infrastructure, machinery, equipment)
- KT0904 Communication and information technology

Internal Assessment Criteria and Weight

- IAC0901 Various resources needed in a workplace are identified and the effect thereof on the output is analysed

(Weight 7%)

4.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 in industry recognised qualifications with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

4.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

5. 252901-001-00-KM-05, Fundamentals of Design Thinking and Innovation, NQF Level 4, Credits 1

5.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of the design thinking principles and application in the workplace

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|-----------------------------------|------------|
| • KM-05-KT01 | : Introduction to design thinking | 15% |
| • KM-05-KT02 | : The human element | 10% |
| • KM-05-KT03 | : Creativity | 20% |
| • KM-05-KT04 | : Innovation | 20% |
| • KM-05-KT05 | : Design | 10% |
| • KM-05-KT06 | : Design thinking methodology | 10% |
| • KM-05-KT07 | : Application of design thinking | 15% |

5.2 Guidelines for Topics

5.2.1 KM-05-KT01 : Introduction to design thinking 15%

Topic elements to be covered include:

- KT0101 Philosophy
- KT0102 Approach and concepts
- KT0103 Definitions and terminology
- KT0104 History

Internal Assessment Criteria and Weight

- IAC0101 Design thinking methodology is understood

(Weight 15%)

5.2.2 KM-05-KT02 : The human element 10%

Topic elements to be covered include:

- KT0201 Human centeredness
- KT0202 Human participation

Internal Assessment Criteria and Weight

- IAC0201 The principles of design thinking is understood

(Weight 10%)

5.2.3 KM-05-KT03 : Creativity 20%

Topic elements to be covered include:

- KT0301 Creativity: is unleashing the potential of the mind to conceive new ideas
- KT0302 Perceiving the world in new ways,
- KT0303 Find hidden patterns,
- KT0304 Make connections between seemingly unrelated phenomena,
- KT0305 Generate solutions
- KT0306 Application in the workplace

Internal Assessment Criteria and Weight

- IAC0301 Creativity is analysed and possible applications are identified

(Weight 20%)

5.2.4 KM-05-KT04 : Innovation

20%

Topic elements to be covered include:

- KT0401 Innovation: is the action of putting things into practical reality, despite challenges and resistance
- KT0402 Different innovation
 - Incremental,
 - Disruptive,
 - Architectural and
 - Radical
- KT0403 Main types of innovation:
 - Process innovation
 - Product innovation
 - Organisational innovation
 - Market innovation
- KT0404 What innovation means to business

Internal Assessment Criteria and Weight

- IAC0401 Innovation is analysed and possible applications are identified

(Weight 20%)

5.2.5 KM-05-KT05 : Design

10%

Topic elements to be covered include:

- KT0501 Think outside the box
- KT0502 Push beyond the obvious solutions

- KT0503 Communication through shape and form

Internal Assessment Criteria and Weight

- IAC0501 Design is analysed and possible applications are identified

(Weight 10%)

5.2.6 KM-05-KT06 : Design thinking methodology

10%

Topic elements to be covered include:

- KT0601 Design thinking phases
- KT0602 Design thinking tools and techniques

Internal Assessment Criteria and Weight

- IAC0601 Design thinking phases, tools and techniques are understood

(Weight 10%)

5.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 in industry recognised qualifications with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

5.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

6. 252901-001-00-KM-06, Refresher Logical Thinking and Basic Calculations, NQF Level 4, Credits 3

6.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to acquire mathematical thinking theory for solving problems and acquire basic maths knowledge for using software toolkits or platforms

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|---|------------|
| • KM-06-KT01 | : Mathematical thinking skills for problem solving. | 15% |
| • KM-06-KT02 | : Basic math | 5% |
| • KM-06-KT03 | : Conversion between decimal and binary systems | 5% |
| • KM-06-KT04 | : Express size and magnitude | 5% |
| • KM-06-KT05 | : Error in calculations | 10% |
| • KM-06-KT06 | : Cartesian coordinate system | 15% |
| • KM-06-KT07 | : Pythagorean theorem for finding the distance between two points | 5% |
| • KM-06-KT08 | : Operator precedence | 10% |
| • KM-06-KT09 | : Integer division | 10% |
| • KM-06-KT10 | : Modulus | 10% |
| • KM-06-KT11 | : Increments | 10% |

6.2 Guidelines for Topics

6.2.1 KM-06-KT01 : Mathematical thinking skills for problem solving. **15%**

Topic elements to be covered include:

- KT0101 Benefits of mathematics.
- KT0102 Mathematical thinking steps for solving problems:
 - Break task down into components
 - Identify similar tasks that may help
 - Identify appropriate knowledge and skills
 - Identify assumptions
 - Select appropriate strategy
 - Consider alternative approaches
 - Look for a pattern or connection
 - Generate examples

Internal Assessment Criteria and Weight

- IAC0101 Problem solving as a complex and reiterative process is reasoned.

(Weight 15%)

6.2.2 KM-06-KT02 : Basic math **5%**

Topic elements to be covered include:

- KT0201 Arithmetic expressions in programming: Addition (+), subtraction (-), multiplication (*) and division (/) for both integers and real data types
- KT0202 Percentage
- KT0203 Negative numbers

Internal Assessment Criteria and Weight

- IAC0201 An understanding of basic math is demonstrated.

(Weight 5%)

6.2.3 KM-06-KT03 : Conversion between decimal and binary systems

5%

Topic elements to be covered include:

- KT0301 Introduction to binary numbers
- KT0302 Perform addition and subtraction of positive whole numbers in binary (16 in decimal, binary and hexadecimal)
- KT0303 Binary arithmetic
- KT0304 Use of BODMAS

Internal Assessment Criteria and Weight

- IAC0301 Conversions between decimal and binary systems and the application thereof are explained.

(Weight 5%)

6.2.4 KM-06-KT04 : Express size and magnitude

5%

Topic elements to be covered include:

- KT0401 Use scientific notation for small and large numbers
- KT0402 Prefixes: From Giga to Pica (10^9 to 10^{-12})

Internal Assessment Criteria and Weight

- IAC0401 Different ways to express size and magnitude is explained.

(Weight 5%)

6.2.5 KM-06-KT05 : Error in calculations

10%

Topic elements to be covered include:

- KT0501 Rational and irrational numbers
- KT0502 Explore repeating decimals and convert them to fraction form
- KT0503 Symbols for irrational numbers
- KT0504 Rounding prematurely in calculations
- KT0505 Accuracy
- KT0506 Final value of a calculation expressed in terms of the required unit

Internal Assessment Criteria and Weight

- IAC0501 The effect of error in calculations is reasoned.

(Weight 10%)

6.2.6 KM-06-KT06 : Cartesian coordinate system

15%

Topic elements to be covered include:

- KT0601 Definition
- KT0602 Terminology
- KT0603 The coordinate plane
- KT0604 Intersecting x- and y-axes
- KT0605 Four quadrants
- KT0606 Naming using Roman numerals
- KT0607 Use and application – creating graphs and maps

Internal Assessment Criteria and Weight

- IAC0601 The Cartesian coordinate system is explained in terms of application and using correct terminology

(Weight 15%)

6.2.7 KM-06-KT07 : Pythagorean theorem for finding the distance between two points

5%

Topic elements to be covered include:

- KT0701 Definition
- KT0702 Terminology
- KT0703 Purpose
- KT0704 Determine the distance between two points on the Cartesian grid.

Internal Assessment Criteria and Weight

- IAC0701 The Pythagorean theorem is explained in terms of application and using correct terminology

(Weight 5%)

6.2.8 KM-06-KT08 : Operator precedence

10%

Topic elements to be covered include:

- KT0801 Definition and terminology
- KT0802 Purpose and use
- KT0803 Order of operations (left to right)
- KT0804 Apply precedence:
 - Parentheses
 - Multiply and divide
 - Add and subtract
- KT0805 Use of PEDMAS in programming

Internal Assessment Criteria and Weight

- IAC0801 Operator precedence is explained in terms of application and using correct terminology

(Weight 10%)

6.2.9 KM-06-KT09 : Integer division

10%

Topic elements to be covered include:

- KT0901 Definition and terminology
- KT0902 Purpose and use
- KT0903 Rules on how to divide integers
- KT0904 Names and symbols for integer division
- KT0905 Software for integer division

Internal Assessment Criteria and Weight

- IAC0901 Integer division is explained in terms of application and using correct terminology and symbols

(Weight 10%)

6.2.10 KM-06-KT10 : Modulus

10%

Topic elements to be covered include:

- KT1001 Definition and terminology
- KT1002 Purpose and use
- KT1003 Modulus abbreviated as “mod”
- KT1004 % application
- KT1005 (==) application
- KT1006 Modulus in programming

Internal Assessment Criteria and Weight

- IAC1001 Modulus is explained in terms of application and using correct terminology and symbols

(Weight 10%)

6.2.11 KM-06-KT11 : Increments

10%

Topic elements to be covered include:

- KT1101 Definition and terminology
- KT1102 Purpose and use
- KT1103 Increment a variable
- KT1104 compound assignment operator
- KT1105 Increments in programming

Internal Assessment Criteria and Weight

- IAC1101 Increments are explained in terms of application and using correct terminology and symbols.

(Weight 10%)

6.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 in industry recognised qualifications with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

6.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

7. 252901-001-00-KM-07, Computers, Devices and Computing Systems, NQF Level 4, Credits 6

7.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of what computers can do, and the processes that make them function in terms of the four major parts: the input, output, CPU (central processing unit), and memory. It gives an overview of networks and connectivity

The learning will enable learners to demonstrate an understanding of:

• KM-07-KT01	: Problem solving skills for IT Professionals	5%
• KM-07-KT02	: Techniques for safety	5%
• KM-07-KT03	: System components	2%
• KM-07-KT04	: Motherboards	2%
• KM-07-KT05	: Processors	3%
• KM-07-KT06	: Memory	2%
• KM-07-KT07	: BIOS and CMOS	5%
• KM-07-KT08	: Hard drives and storage devices	3%
• KM-07-KT09	: Power supplies and voltage	3%
• KM-07-KT10	: Ports, cables, and connectors	2%
• KM-07-KT11	: Input and output devices	2%
• KM-07-KT12	: Installing and managing printers	2%
• KM-07-KT13	: Mobile devices, multimedia, and laptop computers	3%
• KM-07-KT14	: Preventative maintenance	5%
• KM-07-KT15	: Troubleshooting procedures	3%
• KM-07-KT16	: Operating systems	5%
• KM-07-KT17	: Managing files	2%
• KM-07-KT18	: Applications utility, troubleshooting, and optimization	5%
• KM-07-KT19	: Configuring device drivers	8%
• KM-07-KT20	: Networking and wireless connections	3%
• KM-07-KT21	: Recovery	10%
• KM-07-KT22	: Cloud computing	10%
• KM-07-KT23	: Security fundamentals	5%
• KM-07-KT24	: Introduction to applications	5%

7.2 Guidelines for Topics

7.2.1 KM-07-KT-01 : Problem solving skills for IT Professionals 5%

Topic elements to be covered include:

- KT0101 Identification of the problem,
- KT0102 Establishing a probable cause,
- KT0103 Testing to determine the cause,
- KT0104 Establishing a plan to resolve the problem,
- KT0105 Implementing your solution,
- KT0106 Verifying functionality,
- KT0107 Implementing preventative measures,
- KT0108 Documenting results

Internal Assessment Criteria and Weight

- IAC0101 Problem-solving steps and the application thereof are explained.

(Weight 5%)

7.2.2 KM-07-KT-02 : Techniques for safety

5%

Topic elements to be covered include:

- KT0201 Personal protective equipment,
- KT0202 Chemicals
- KT0203 Cables
- KT0204 Hazards (Tripping, electrical, fire, jewellery, etc.)
- KT0205 Environmental risks
- KT0206 Ergonomics and lifting techniques

Internal Assessment Criteria and Weight

- IAC0201 Proper techniques for handling computer components for the safety and the longevity of the hardware are identified and described.

(Weight 5%)

7.2.3 KM-07-KT-03 : System components

2%

Topic elements to be covered include:

- KT0301 Storage devices such as hard drives, magnetic tapes, flash drives.
- KT0302 Solid-state drives, such as IDE, SATA, and SCSI.
- KT0303 Processors and CPUs, processing speeds, operating systems, and controls.

Internal Assessment Criteria and Weight

- IAC0301 Various types of system components are identified and described

(Weight 2%)

7.2.4 KM-07-KT-04 : Motherboards

2%

Topic elements to be covered include:

- KT0401 Function of motherboards.
- KT0402 Location of primary components.
- KT0403 Motherboard chipsets, audio connections, drive activity light, the power light, on/off buttons, and USB connections.

Internal Assessment Criteria and Weight

- IAC0401 Understanding of motherboards, primary components and related aspects is demonstrated.

(Weight 2%)

7.2.5 KM-07-KT-05 : Processors **3%**

Topic elements to be covered include:

- KT0501 Processors.
- KT0502 Terminology.
- KT0503 Processor speeds in megahertz and gigahertz.
- KT0504 Architecture, including registers, cache memory, virtualization, graphics processing units, throttling, and overclocking.

Internal Assessment Criteria and Weight

- IAC0501 Understanding of processors and related aspects is demonstrated.

(Weight 3%)

7.2.6 KM-07-KT-06 : Memory **2%**

Topic elements to be covered include:

- KT0601 Types of memory: RAM and ROM including DRAM, SRAM, CMOS RAM, and VRAM.
- KT0602 Purpose of memory.

Internal Assessment Criteria and Weight

- IAC0601 Purpose of memory and how it relates to computer components is described.

(Weight 2%)

7.2.7 KM-07-KT-07 : BIOS and CMOS **5%**

Topic elements to be covered include:

- KT0701 BIOS: Basic Input-Output System that runs a computer.
- KT0702 Upgrade BIOS
- KT0703 Potential issues
- KT0704 CMOS: Complementary Metal Oxide Semiconductor
- KT0705 CMOS setup, settings, and access

Internal Assessment Criteria and Weight

- IAC0701 Understanding of BIOS and CMOS operations is demonstrated.

(Weight 5%)

7.2.8 KM-07-KT-08 : Hard drives and storage devices **3%**

Topic elements to be covered include:

- KT0801 Hard drives and storage devices
- KT0802 Installing discs, formatting discs, partitioning discs, troubleshooting disks, as well as IDE, SCSI, and SATA.
- KT0803 Parts of a hard disk, such as platters, tracks, sectors, clusters, and cylinders.
- KT0804 Installing IDE devices

Internal Assessment Criteria and Weight

- IAC0801 An understanding of hard drives and storage devices and components is demonstrated.
- IAC0802 Installing procedures are explained.

(Weight 3%)

7.2.9 KM-07-KT-09 : Power supplies and voltage

3%

Topic elements to be covered include:

- KT0901 Power supply units, uninterruptible power supplies.
- KT0902 Identifying power supplies.
- KT0903 Amps, ohms, volts, and watts
- KT0904 Affecting system hardware and components.

Internal Assessment Criteria and Weight

- IAC0901 Power supply and the effect on system hardware and components are explained.

(Weight 3%)

7.2.10 KM-07-KT-10 : Ports, cables, and connectors

2%

Topic elements to be covered include:

- KT1001 Ports, cables, and connectors
- KT1002 Power supply units, uninterruptible power supplies, and identifying power supplies.
- KT1003 Conversion of voltage, overcurrent protection, input supply voltage, DC voltage regulation, input under-voltage, and Energy Star designation.

Internal Assessment Criteria and Weight

- IAC1001 An understanding of ports, cables, and connectors and the effect of voltage are demonstrated.

(Weight 2%)

7.2.11 KM-07-KT-11 : Input and output devices

2%

Topic elements to be covered include:

- KT1101 Input devices and function
- KT1102 Output devices and function
- KT1103 Keyboards, mice, and touchscreens

Internal Assessment Criteria and Weight

- IAC1101 Input and output devices are identified and the respective functions are explained.

(Weight 2%)

7.2.12 KM-07-KT-12 : Installing and managing printers **2%**

Topic elements to be covered include:

- KT1201 Types of printers
- KT1202 Installing and managing printers
- KT1203 Drivers, paper feed, networking, and user error

Internal Assessment Criteria and Weight

- IAC1201 An understanding of types of printers, installing printers and dealing with problems is demonstrated.

(Weight 2%)

7.2.13 KM-07-KT-13 : Mobile devices, multimedia, and laptop computers **3%**

Topic elements to be covered include:

- KT1301 Mobile devices,
- KT1302 Multimedia,
- KT1303 Laptop computers
- KT1304 Troubleshooting common problems: battery or power supply issues, verifying connections, checking the wireless switch, and verifying software configuration

Internal Assessment Criteria and Weight

- IAC1301 The respective functions of mobile devices, multimedia, and laptop computers are identified.
- IAC1302 Common problems are listed.
- IAC1303 Troubleshooting procedures are explained.

(Weight 3%)

7.2.14 KM-07-KT-14 : Preventative maintenance **5%**

Topic elements to be covered include:

- KT1401 Purpose of preventive maintenance
- KT1402 Benefits
- KT1403 Prevent data loss
- KT1404 Improve performance
- KT1405 Preventive tasks: e.g. Updating virus and malware detectors, backing up data, defragmenting drives, etc.

Internal Assessment Criteria and Weight

- IAC1401 The importance of preventative maintenance is justified.
- IAC1402 The purpose and benefits of preventative maintenance is explained.

- IAC1403 Preventative procedures are explained.

(Weight 5%)

7.2.15 KM-07-KT-15 : Troubleshooting procedures

3%

Topic elements to be covered include:

- KT1501 Troubleshooting process
- KT1502 Hardware tools required
- KT1503 Software tools e.g. boot disks and system rescue, etc.

Internal Assessment Criteria and Weight

- IAC1501 Troubleshooting procedures are explained.
- IAC1502 Hardware and software tools and the respective functions are described.

(Weight 3%)

7.2.16 KM-07-KT-16 : Operating systems

5%

Topic elements to be covered include:

- KT1601 Functions of an operating system: e.g. managing hardware, resources, data, and making user interaction easy through the use of a graphical user interface (GUI)
- KT1602 Big three operating systems: Windows, Linux, and Mac.
- KT1603 Control panel
- KT1604 Remote assistance
- KT1605 OS command prompt
- KT1606 OS migration
- KT1607 OS configurations

Internal Assessment Criteria and Weight

- IAC1601 An understanding of operating systems is demonstrated.

(Weight 5%)

7.2.17 KM-07-KT-17 : Managing files

2%

Topic elements to be covered include:

- KT1701 Files and directories, naming conventions, and the File Allocation Table (FAT)
- KT1702 Common executable file extensions, such as batch files, command files, and power shell scripts and purpose

Internal Assessment Criteria and Weight

- IAC1701 Types of files and the respective purposes are identified.

(Weight 2%)

7.2.18 KM-07-KT-18 : Applications utility, troubleshooting, and optimization

5%

Topic elements to be covered include:

- KT1801 Upgrade and remove applications from computers.
- KT1802 New installs through disc or internet download.
- KT1803 Common processes for downloading and removing software.
- KT1804 Troubleshooting
- KT1805 Optimisation

Internal Assessment Criteria and Weight

- IAC1801 Procedures for installing and removing utilities are described.

(Weight 5%)

7.2.19 KM-07-KT-19 : Configuring device drivers

8%

Topic elements to be covered include:

- KT1901 Configuring device drivers
- KT1902 Changing device settings
- KT1903 Updating drivers
- KT1904 Removing device drivers
- KT1905 Plug-and-play devices
- KT1906 Device failure
- KT1907 System restore points

Internal Assessment Criteria and Weight

- IAC1901 Considerations for configuring device drivers are reasoned.

(Weight 8%)

7.2.20 KM-07-KT-20 : Networking and wireless connections

3%

Topic elements to be covered include:

- KT2001 Peer-to-peer networks and server networks,
- KT2002 Common functionality of server networks,
- KT2003 Three types of network topology: Bus, Star, and Ring networks
- KT2004 Networking devices including network interface cards, repeaters, bridges, routers, gateways, hubs, and switches

Internal Assessment Criteria and Weight

- IAC2001 Networking and wireless connections are discussed with reference to types and functionality.

(Weight 3%)

7.2.21 KM-07-KT-21 : Recovery

10%

Topic elements to be covered include:

- KT2101 Recovering systems and disaster recovery
- KT2102 Process for disaster recovery
- KT2103 Expecting things to go wrong
- KT2104 Using boot logs to troubleshoot problems
- KT2105 Process for booting a system into safe mode
- KT2106 Emergency repair
- KT2107 Factory repair partitions
- KT2108 Startup repair
- KT2109 System image recovery

Internal Assessment Criteria and Weight

- IAC2101 Recovery processes are identified and application is described.

(Weight 10%)

7.2.22 KM-07-KT-22 : Cloud computing

10%

Topic elements to be covered include:

- KT2201 Network or internet connection
- KT2202 Manipulate, collate, configure, and access applications online
- KT2203 Providing support infrastructure, applications, and data storage
- KT2204 Benefits and risks to cloud computing as well as cloud characteristics
- KT2205 Cloud storage

Internal Assessment Criteria and Weight

- IAC2201 Cloud computing functionality, benefits and risks are identified and described.

(Weight 10%)

7.2.23 KM-07-KT-23 : Security fundamentals

5%

Topic elements to be covered include:

- KT2301 Business continuity,
- KT2302 Disaster recovery planning,
- KT2303 Application development security,
- KT2304 Access control,
- KT2305 Information security governance,
- KT2306 Risk management,
- KT2307 Operation security,
- KT2308 Physical security,

- KT2309 Network security

Internal Assessment Criteria and Weight

- IAC2301 Typical security risks are listed.
- IAC2302 The importance of counter measures is justified.

(Weight 5%)

7.2.24 KM-07-KT-24 : Introduction to applications

5%

Topic elements to be covered include:

- KT2401 Software development
- KT2402 Game development
- KT2403 Software tools
- KT2404 Mobile Apps
- KT2405 Internet of things

Internal Assessment Criteria and Weight

- IAC2401 Various software developments are listed.

(Weight 5%)

7.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 in industry recognised qualifications with 1 years' experience in the IT industry
 - CYBERSECURITY vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training

- OHS compliance certificate
- Ethical clearance (where necessary)

7.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

8. 252901-001-00-KM-08, Data and Databases Vulnerabilities, NQF Level 4, Credits 3

8.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of data and databases and giving meaning to data through data processing, analysis and visualisation

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|--|------------|
| • KM-08-KT01 | : Data vulnerability and security | 15% |
| • KM-08-KT02 | : Data and data processing | 15% |
| • KM-08-KT03 | : Databases, data storage and access to data | 20% |
| • KM-08-KT04 | : Structured query language (SQL) | 10% |
| • KM-08-KT05 | : Data scraping | 20% |
| • KM-08-KT06 | : Software for analysing and visualising data. | 20% |

8.2 Guidelines for Topics

8.2.1 KM-08-KT01 : Data vulnerability and security **15%**

Topic elements to be covered include:

- KT0101 Definition
- KT0102 Data vulnerability, risk and exploitation
- KT0103 Stages of data vulnerability: at network level, at system level, at data level.
- KT0104 Purpose of protecting data
- KT0105 Process for protecting data
- KT0106 Unauthorised access, elevation of privileges or denial of data
- KT0107 Data corruption
- KT0108 Data security solutions

Internal Assessment Criteria and Weight

IAC0101 The importance of data security is reasoned.

(Weight 15%)

8.2.2 KM-08-KT02 : Data and data processing **15%**

Topic elements to be covered include:

- KT0201 Value of data
- KT0202 Data analysis for cybersecurity: Importance of analysis
- KT0203 Data sourcing:
 - Data sources
 - Data types
 - Reliable data
 - Automated data collection
- KT0204 Refining data:

- Missing data
- Data misalignments
- Separating useful data from the rest
- KT0205 Flaws in data:
 - Commission
 - Omission
 - Perspective
 - Bias
 - Frame of reference
- KT0206 Limits of data acquisition
- KT0207 Data:
 - Setting up data,
 - Data interactions,
 - Assigned to different fields

Internal Assessment Criteria and Weight

- IAC0201 Data and data processing principles are understood and explained

(Weight 15%)

8.2.3 KM-08-KT03 : Databases, data storage and access to data

20%

Topic elements to be covered include:

- KT0301 Database:
 - definition, components, function, types, characteristics of good databases, structure and challenges
- KT0302 Data:
 - storage, characteristics of quality data,
- KT0303 Data:
 - collection warehousing, mining and managing concepts
- KT0304 Relational database design
- KT0305 Database design tools
- KT0306 Create, design and modify relational database
- KT0307 Import and export data
- KT0308 Design and create queries

Internal Assessment Criteria and Weight

- IAC0301 Databases concepts, principles and characteristics are described.
- IAC0302 Data concepts, principles and characteristics are described.
- IAC0303 Database design concepts, principles and tools are described.
- IAC0304 Access to data is described.

(Weight 20%)

8.2.4 KM-08-KT04 : Structured query language (SQL)

10%

Topic elements to be covered include:

- KT0401 SQL programming language
- KT0402 SQL code constructs to perform database transactions
- KT0403 Storing, retrieving, managing or manipulating the data inside a relational database management system (RDBMS)

Internal Assessment Criteria and Weight

- IAC0401 The application of SQL is explained.

(Weight 10%)

8.2.5 KM-08-KT05 : Data scraping

20%

Topic elements to be covered include:

- KT0501 Concept and definition
- KT0502 Purpose of data scraping
- KT0503 Data scraping tools
- KT0504 Legal issues
- KT0505 Web scraping procedure:
 - Find the URL to scrape
 - Inspecting the page
 - Find the data you want to extract
 - Write the code
 - Run the code and extract the data
 - Store the data in the required format
- KT0506 Libraries used for web scraping

Internal Assessment Criteria and Weight

- IAC0501 The principles and purpose of data scraping are explained.

(Weight 20%)

8.2.6 KM-08-KT06 : Software for analysing and visualising data.

20%

Topic elements to be covered include:

- KT0601 Reporting
- KT0602 Tables
- KT0603 Pivot tables and pivot charts
- KT0604 Dashboards
- KT0605 Hierarchies and time data
- KT0606 The data model
- KT0607 Importing data from files
- KT0608 Importing data from databases
- KT0609 Importing data from reports
- KT0610 Creating and formatting measures
- KT0611 Visualizing data

Internal Assessment Criteria and Weight

- IAC0601 The importance of analysing and visualising data is reasoned.

(Weight 20%)

8.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 in industry recognised qualifications with 1 years' experience in the IT industry
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

8.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

9. 252901-001-00-KM-09, Introduction to 4IR and Future Skills, NQF Level 4, Credits 4

9.1 Purpose of the Knowledge Module

The main focus of the learning in this knowledge module is to build an understanding of the impact of 4IR on communities, individuals and businesses and important skills for future needs

The learning will enable learners to demonstrate an understanding of:

- | | | |
|--------------|--|-----|
| • KM-09-KT01 | : 4 IR emerging trends | 10% |
| • KM-09-KT02 | : Computing Knowledge | 7% |
| • KM-09-KT03 | : Future skills and competencies (4IR) | 10% |
| • KM-09-KT04 | : 4 IR trends affecting businesses | 10% |
| • KM-09-KT05 | : Interpersonal skills | 5% |
| • KM-09-KT06 | : Intrapersonal skills | 5% |
| • KM-09-KT07 | : Communication principles and methods | 5% |
| • KM-09-KT08 | : Written business communication | 7% |
| • KM-09-KT09 | : Presentation skills | 7% |
| • KM-09-KT10 | : Teamwork in the workplace | 10% |
| • KM-09-KT11 | : Committees and meetings | 5% |
| • KM-09-KT09 | : Job descriptions and profiles | 5% |
| • KM-09-KT13 | : Customers and stakeholders | 7% |
| • KM-09-KT14 | : Customer service | 7% |

9.2 Guidelines for Topics

9.2.1 KM-09-KT01 : 4 IR emerging trends 10%

Topic elements to be covered include:

- KT0101 Artificial intelligence
- KT0102 Cloud computing
- KT0103 Cyber security
- KT0104 Data science
- KT0105 Internet of things
- KT0106 Quality engineering automation
- KT0107 Robotic processing automation
- KT0108 Software programming
- KT0109 Design thinking and innovation
- KT0110 e-Waste

Internal Assessment Criteria and Weight

- IAC0101 The transformational effect that 4IR is having or may have on business ecosystems is understood through a comprehensive knowledge of the emerging trends within 4IR

(Weight 10%)

9.2.2 KM-09-KT02 : Computing Knowledge 7%

Topic elements to be covered include:

- KT0201 Introduction to programming language
- KT0202 Programming basics
- KT0203 Basic programming knowledge on HTML, JavaScript (or any scripting language)

- KT0204 Software development, e.g. C#, C++, Java, .NET
- KT0205 Databases (SQL or NoSQL)
- KT0206 Web development technologies

Internal Assessment Criteria and Weight

- IAC0201 A sound knowledge of the basics of programming and software development, including various development languages, databases and web development, is demonstrated

(Weight 7%)

9.2.3 KM-09-KT03 : Future skills and competencies (4IR)

10%

Topic elements to be covered include:

- KT0301 Disruptive thinking (encourage this) (application to their own environment)
- KT0302 Continuously searching for ideas
- KT0303 Think innovatively (analyse the current market and come up with solutions to the current problems)
- KT0304 Soft skills
- KT0305 Programming languages
- KT0306 Operating systems
- KT0307 Open source
- KT0308 Tools for a cloud environment (for configuration and management), tools for debugging, login and monitoring and tools for image
- KT0309 Familiarity with Office tools
- KT0310 Leadership and people management skills

Internal Assessment Criteria and Weight

- IAC0301 A sound knowledge of programming languages, operating systems, the cloud environment, Office tools, etc. is demonstrated
- IAC0302 The importance of analytical and innovative thinking within a business ecosystem that encompasses 4IR, is understood
- IAC0303 Soft skills as well as leadership and people management skills are understood and applied

(Weight 10%)

9.2.4 KM-09-KT04 : 4 IR trends affecting businesses

10%

Topic elements to be covered include:

- KT0401 Afro-centric approach to African problems – taking the best from the existing products and coming up with own solutions Continental challenges and opportunities
- KT0402 Using Google, Amazon and MS forms and tools to reduce development time (e.g. embed AI APIs)
- KT0403 Business intelligence applications and availability of Big Data (collecting data, converting data into information and turning information into knowledge, knowledge into intelligence and intelligence into wisdom)
- KT0404 Collecting data on clients
- KT0405 Insight into different markets
- KT0406 Automated factories

- KT0407 Exposure to the global world

Internal Assessment Criteria and Weight

- IAC0401 The various IR trends affecting businesses are described and understood within an Afro-centric context, taking into account the existing continental challenges and opportunities for development
- IAC0402 Business Intelligence and Big Data are applied and utilized as marketing tool
- IAC0403 Knowledge of automated factories and insight into the different global markets are displayed

(Weight 10%)

9.2.5 KM-09-KT05 : Interpersonal skills

5%

Topic elements to be covered include:

- KT0501 Concept, definition and terminology
- KT0502 Principles
- KT0503 Attributes:
 - Social intelligence
 - Confidentiality
 - Conflict handling and resolution
 - Decision making
 - Defending vs attacking
 - Problem solving, Troubleshooting
 - Respect
 - Roles, responsibilities
 - Thinking about the end-user

Internal Assessment Criteria and Weight

- IAC0501 The following universal soft skills are applied in an IR report: flexibility, communication skills, teamwork, literacy, emotional maturity, decision making and adaptability

(Weight 5%)

9.2.6 KM-09-KT06 : Intrapersonal skills

5%

Topic elements to be covered include:

- KT0601 Concept, definition and terminology
- KT0602 Principles
- KT0603 Attributes:
 - Adaptability
 - Agility
 - Analytical thinking
 - Cognitive thinking skills
 - Emotional maturity
 - Flexibility
 - Planning
 - Problem solving

- Reflection
- Research and investigate
- Self-management
- Strong attention to detail
- Time-management
- Resilience

Internal Assessment Criteria and Weight

- IAC0601 The concept and principles of the intrapersonal skills needed to function effectively within a business ecosystem and with the demands of 4IR in mind, are understood
- IAC0602 The importance of an ability to solve problems through planning, research, investigation, analytical thinking and strong attention to detail, is described
- IAC0603 Effective self-management and time-management are applied

(Weight 5%)

9.2.7 KM-09-KT07 : Communication principles and methods

5%

Topic elements to be covered include:

- KT0701 Concept, definition and terminology
- KT0702 The different types and forms of communication and communication processes
- KT0703 Communication methods
- KT0704 Barriers to communication
- KT0705 Communication network: Interdepartmental, Supply chain network, etc.
- KT0706 Advantages of good communication
- KT0707 Consequences of poor/no communication

Internal Assessment Criteria and Weight

- IAC0701 The basic principles of communication within a business ecosystem are explained
- IAC0702 Types of communication are defined and compared
- IAC0703 Various communication methods and techniques are compared for effectiveness in the workplace
- IAC0704 Communication flow in the workplace is evaluated
- IAC0705 Reporting structures in the workplace are evaluated
- IAC0706 The importance and the role of recording and reporting of data and information are explained

(Weight 5%)

9.2.8 KM-09-KT08 : Written business communication

7%

Topic elements to be covered include:

- KT0801 Business requirement specifications
- KT0802 Types

- KT0803 Conventions
- KT0804 Schedules
- KT0805 Reports, reporting protocols and methods
- KT0806 Manuals and guidelines
- KT0807 Work instructions/briefs
- KT0808 Technical report writing
- KT0809 Extracting information from written texts
- KT0810 Policies aligned to standard (IEEE 829-2008 standards)

Internal Assessment Criteria and Weight

- IAC0801 The purpose of written communication within a business ecosystem is stated
- IAC0802 Effective written communication is evaluated
- IAC0803 Work instructions are interpreted and applied
- IAC0804 The purpose and advantages of communication are described
- IAC0805 Definition and purpose of work instructions are described

(Weight 7%)

9.2.9 KM-09-KT09 : Presentation skills

7%

Topic elements to be covered include:

- KT0901 Concept, definition and terminology
- KT0902 Types: visual, verbal, written
- KT0903 Conventions
- KT0904 Presenting options and solutions
- KT0905 Presenting technical details
- KT0906 Visualisation of business intelligence
- KT0907 Suitable APIs and storytelling using the right tools to:
 - present,
 - frame the story,
 - focus on certain aspects,
 - pitch,
 - clear terms,
 - pictorial

Internal Assessment Criteria and Weight

- IAC0901 The definition and importance of effective presentation skills are understood
- IAC0902 Effective presentation, either visually, verbally or in written form, is applied
- IAC0903 The technical aspects of and solutions for effective presenting, are described
- IAC0904 Suitable APIs are selected and applied in the form of stories in order to visually present business intelligence within the business

(Weight 7%)

9.2.10 KM-09-KT10 : Teamwork in the workplace

10%

Topic elements to be covered include:

- KT1001 Concept, definition and terminology
- KT1002 Principles of teamwork
- KT1003 Advantages of teamwork
- KT1004 Team composition and members
- KT1005 Roles, responsibilities and functions
- KT1006 Team dynamics
- KT1007 Common goals and collaboration
- KT1008 Nature of multidisciplinary teams and teamwork
- KT1009 Setting and achieving targets
- KT1010 Collaboration tools (electronic)

Internal Assessment Criteria and Weight

- IAC1001 The composition of a team within a business ecosystem is described
- IAC1002 The role of a team and its various members is described
- IAC1003 The contribution of a team to an organisation is evaluated
- IAC1004 The dynamics of a well-functioning team are described
- IAC1005 The influence of different team members on team performance is analysed
- IAC1006 External factors which will impact on teamwork are defined
- IAC1007 Internal factors which will impact on teamwork are defined

(Weight 10%)

9.2.11 KM-09-KT11 : Committees and meetings

5%

Topic elements to be covered include:

- KT1101 Procedures
- KT1102 Agendas and minutes
- KT1103 Roles and responsibilities
- KT1104 WSP committees
- KT1105 EE committees
- KT1106 Safety and health committees
- KT1107 Wellness committees

Internal Assessment Criteria and Weight

- IAC1101 The roles and functions of the various committees are defined
- IAC1102 The purpose and objectives of meetings are defined
- IAC1103 Input on correctness of minutes is constructed

(Weight 5%)

9.2.09 KM-09-KT12 : Job descriptions and profiles

5%

Topic elements to be covered include:

- KT1201 Purpose
- KT1202 Job and person specification
- KT1203 Content
- KT1204 Alignment to performance standards

Internal Assessment Criteria and Weight

- IAC1201 Definition and purpose of a job description is described
- IAC1202 A job description is compiled
- IAC1203 A job profile is compiled

(Weight 5%)

9.2.13 KM-09-KT13 : Customers and stakeholders

7%

Topic elements to be covered include:

- KT1301 Concept, definition and terminology
- KT1302 Types of customers
- KT1303 Customer profile
- KT1304 Typical customer behaviour: including habits and mannerisms
- KT1305 Difficult customers
- KT1306 Customer care
- KT1307 Stakeholder management and participation

Internal Assessment Criteria and Weight

- IAC1301 The importance of knowledge about the various types of customers and stakeholders operating within a business ecosystem, is understood
- IAC1302 Culture, habits and nature of customers and stakeholders are taken into account when interaction with these parties is necessitated

(Weight 7%)

9.2.14 KM-09-KT14 : Customer service

7%

Topic elements to be covered include:

- KT1401 Concept, definition and terminology
- KT1402 Customer service principles
- KT1403 Customer centeredness
- KT1404 Handover and sign-off procedures and techniques

- KT1405 Technical documentation
- KT1406 Training in the use of the system

Internal Assessment Criteria and Weight

- IAC1401 The definition, principles and importance of customer service are understood and described
- IAC1402 Technical aspects of customer service, including sign-off procedures and techniques and knowledge of the system used, are described and applied

(Weight 7%)

9.3 Provider Programme Accreditation Criteria

Physical Requirements:

- The provider must have lesson plans and structured learning material or provide learners with access to structured learning material that addresses all the topics in all the knowledge modules as well as the applied knowledge in the practical skills.
- QCTO/ MICT SETA requirements

Human Resource Requirements:

- Lecturer/learner ratio of 1:20 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 in industry recognised qualifications with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

9.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

SECTION 3B: PRACTICAL SKILLS MODULE SPECIFICATIONS

List of Practical Skills Module Specifications

Practical Skills	252901-001-00-PM-01	Ensure Compliance in terms of Legal Cybersecurity Requirements and National and International Standards	5	4
Practical Skills	252901-001-00-PM-02	Assess Risks and Vulnerabilities and Evaluate Current Security Measures	5	20
Practical Skills	252901-001-00-PM-03	Implement Protection, Prevention and Detection Measures to Mitigate Risk, Violations and Vulnerabilities	5	20
Practical Skills	252901-001-00-PM-04	Apply Logical Thinking and Maths	4	6
Practical Skills	252901-001-00-PM-05	Apply Basic Scriptwriting for Cybersecurity Toolsets	4	4
Practical Skills	252901-001-00-PM-06	Access and Visualise Structured Data Using Spreadsheets	4	5
Practical Skills	252901-001-00-PM-07	Applying Design Thinking Methodologies	4	4
Practical Skills	252901-001-00-PM-08	Function Ethically and Effectively as a Member of a Multidisciplinary Team	4	5

1. 252901-001-00-PM-01, Ensure Compliance in terms of Legal Cybersecurity Requirements and National and International Standards, NQF Level 5, Credits 4

1.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to participate in organisational governance ensuring all systems are put in place to achieve compliance with legal requirements and nationally accepted standards. In addition they will acquire skills to configure computers and mobile devices to connect to networks and to the internet

The learner will be required to:

- PM-01-PS01 : Ensure compliance of computer systems, networks and data with legal cybersecurity requirements and national and international standards
- PM-01-PS02 : Apply basic computing and IT infrastructure procedures to ensure secure computer use
- PM-01-PS03 : Set up computer systems and networks applying applicable cybersecurity protocols
- PM-01-PS04 : Protect, detect, and respond to network attacks as first line defence

1.2 Guidelines for Practical Skills

- 1.2.1 PM-01-PS01 : Ensure compliance of computer systems, networks and data with legal cybersecurity requirements and national and international standards**

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform and a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0101 Implement the security controls as stipulated in the cybersecurity framework
- PA0102 Comment and advise on cybersecurity related standards and company compliance when needed
- PA0103 Draft a cybersecurity policy which is legally compliant
- PA0104 Ensure cybersecurity compliance of systems, networks and data
- PA0105 Participate in organisational governance and compliance by ensuring all systems are put in place to achieve the objectives and key performance indicators (KPI's) of the business

Applied Knowledge

- AK0101 Legal, policy requirements, national and international standards
- AK0102 Consequences of non-compliance

Internal Assessment Criteria

- IAC0101 Security controls are implemented and are compliant with the cybersecurity framework
- IAC0102 Cybersecurity compliance with company standards, systems, networks and data is ensured
- IAC0103 A legal cybersecurity policy is drafted
- IAC0104 Achievement of objectives and KPI's is ensured by participation in organizational governance

- 1.2.2 PM-01-PS02 : Apply basic computing and IT infrastructure procedures to ensure secure computer use**

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0201 Identify components of laptop and network and connect to a network according to company cybersecurity policies
- PA0202 Identify identity theft, phishing scams, malware, social engineering, and financial frauds
- PA0203 Safeguard mobile and media and protect data
- PA0204 Protect computers, accounts, and social networking profiles
- PA0205 Secure operating systems (OS)
- PA0206 Identify malware
- PA0207 Install antivirus and clear a virus
- PA0208 Ensure internet security
- PA0209 Ensure security on social networking sites
- PA0210 Ensure security of email communications
- PA0211 Ensure security of mobile devices
- PA0212 Ensure security of the cloud
- PA0213 Ensure security of network connections
- PA0214 Provide data backup to ensure disaster recovery
- PA0215 Report regularly on incidents, breaches, risks and vulnerabilities, including lessons learnt and recommendations for continuous improvements

Applied Knowledge

- AK0201 Legal, policy requirements, national and international standards
- AK0202 Introduction to security
- AK0203 Fundamentals of various computer and network security threats
- AK0204 Understanding of security incidents and reporting

Internal Assessment Criteria

- IAC0201 Laptop components and connection to network according to cybersecurity policies are understood
- IAC0202 Various possible threats to operating systems, social networking platforms and connections, accounts, communication systems, data and all computing devices are identified and the necessary protection is installed
- IAC0203 Back-ups are made on a regular basis and incidents, breaches, risks and vulnerabilities are reported
- IAC0204 Recommendations for improvements are made

1.2.3 PM-01-PS03 : Set up computer systems and networks applying applicable cybersecurity protocols

Scope of Practical Skill

Given access to a cybersecurity toolkit, a virtual environment suitable to the toolkit/platform and scenario, the learner must be able to:

- PA0301 Identify components of laptop and network and connect to a network according to company cybersecurity policies
- PA0302 Install and configure toolkit/software
- PA0303 Support operating systems
- PA0304 Install and configure PC system unit component and peripheral devices
- PA0305 Install and configure storage devices and troubleshoot
- PA0306 Install and configure internal system components and troubleshoot
- PA0307 Install and configure operating systems, troubleshoot and maintain
- PA0308 Identify all the components of the network
- PA0309 Set up and troubleshoot IP addresses

- PA0310 Configure and troubleshoot network connections
- PA0311 Set up SSID, wireless access points, etc. (set it up securely, apply applicable security protocols)
- PA0312 Manage users, workstations and shared resources
- PA0313 Implement client virtualisation and cloud computing
- PA0314 Implement physical security
- PA0315 Secure workstations (PC, laptops) and data and troubleshoot security issues
- PA0316 Install, configure, troubleshoot shared folders and printers with secure group access and connection to prevent illegal access

Applied Knowledge

- AK0301 Network infrastructure concepts

Internal Assessment Criteria

- IAC0301 Computer and network components are identified and connected to secure network
- IAC0302 Storage devices, internal system components, operating systems, IP addresses and network connections are installed and configured and secured by troubleshooting
- IAC0303 SSID, wireless access points, etc. are set up and users, work stations and shared resources are managed
- IAC0304 Client virtualisation, cloud computing and physical security are implemented
- IAC0305 Workstations and data are secured and any security issues handled through troubleshooting
- IAC0306 Shared folders and printers are installed and configured and secure group connection and access are ensured by troubleshooting

1.2.4 PM-01-PS04 : Protect, detect, and respond to network attacks as first line defence

Scope of Practical Skill

Given access to a cybersecurity toolkit, a virtual environment suitable to the toolkit/platform and scenario, the learner must be able to:

- PA0401 Apply relevant toolkit/platform
- PA0402 Apply network security controls, protocols, perimeter appliances, secure IDS's, VPN's, and firewall configuration and identify risk
- PA0403 Use devices such as switches and routers to segment network traffic and create resilient networks
- PA0404 Secure access to central services, and identify the risks
- PA0405 Conduct network traffic signature, analysis, and vulnerability scanning
- PA0406 Identify benefits and drawbacks of existing network configurations
- PA0407 Identify network security threats, vulnerabilities, and attacks
- PA0408 Implement network security controls, protocols, and devices
- PA0409 Design and implement network security policy
- PA0410 Ensure physical security
- PA0411 Ensure host security
- PA0412 Configure secure firewall protection
- PA0413 Secure IDS configuration and troubleshoot
- PA0414 Ensure secure VPN configuration and troubleshoot
- PA0415 Set up wireless network defence
- PA0416 Monitor and analyse network traffic
- PA0417 Monitor network risk and vulnerability

- PA0418 Troubleshoot network problems using network troubleshooting methodologies and appropriate tools to support connectivity and performance
- PA0419 Monitor data backup to ensure recovery
- PA0420 Execute network incident response
- PA0421 Report regularly on incidents, breaches, risks, vulnerabilities and cybersecurity state of the system, including lessons learnt and recommendations for continuous improvements

Applied Knowledge

- AK0401 Network defence fundamentals
- AK0402 Details of network traffic signature, analysis, and vulnerability scanning

Internal Assessment Criteria

- IAC0401 Various network defence devices are applied and risks identified
- IAC0402 Switches and routers are used to segment network traffic and create resilient networks
- IAC0403 Existing network configurations are investigated and threats and vulnerabilities are identified
- IAC0404 Network security controls, protocols, devices and policy are designed and implemented
- IAC0405 Physical and host security are ensured
- IAC0406 Secure IDS and VPN configurations are ensured using troubleshooting methods
- IAC0407 Wireless network defences are set up and network traffic is monitored, taking risks and vulnerabilities into account
- IAC0408 Data backup is monitored Regular reports on breaches, vulnerabilities and level of cybersecurity of the system are done and recommendations on improvements made

1.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 industry recognised qualification with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

1.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

2. 252901-001-00-PM-02, Assess Risks and Vulnerabilities and Evaluate Current Security Measures, NQF Level 5, Credits 20

2.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to assess risks, vulnerabilities and trends and evaluate current security measures and security posture of an organisation

The learner will be required to:

- PM-02-PS01 : Apply and use functions of a suitable toolset for cybersecurity purposes
- PM-02-PS02 : Scan systems for and identify vulnerabilities and potential security threats to information systems and analyse their consequences
- PM-02-PS03 : Evaluate the level of risk in an IT system using standard security models and apply appropriate countermeasures
- PM-02-PS04 : Use appropriate tools to assess the security posture of an organisation

2.2 Guidelines for Practical Skills

2.2.1 PM-02-PS01 : Apply and use functions of a suitable toolset for cybersecurity purposes

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0101 Access and apply the various functions of the specific platform.

Applied Knowledge

- AK0101 Examples of tools (examples and not exhaustive list): Nessus, OpenVAS, Kali Linux, BackTrack, Wireshark, Snort, Nmap, Splunk, ArchSight, Symantec, IDS/IPS, Cisco CheckPoint, and Request Tracker for Incident Response (RTIR), Malware Information Sharing Platform (MISP), FortiGate

Internal Assessment Criteria

- IAC0101 Demonstrated ability to apply and use functions of a suitable toolset for cybersecurity purposes

2.2.2 PM-02-PS02 : Scan systems for and identify vulnerabilities and potential security threats to information systems and analyse their consequences

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0201 Identify the business risk by conducting brainstorming session and/or research
- PA0202 Put risk catalogue together
- PA0203 Passively test security controls
- PA0204 Identify vulnerability
- PA0205 Identify lack of security controls
- PA0206 Identify common misconfigurations
- PA0207 Put systems into place ensuring that only the right people have access to systems and data and do penetration testing to determine whether they are sound

- PA0208 Investigate whether other companies experienced similar situation and what lessons were learnt (Use this as a starting point).
- PA0209 Find the best way to secure the IT infrastructure of an organisation

Applied Knowledge

- AK0201 Legal, policy requirements, national and international standards
- AK0202 Intrusive vs. non-intrusive
- AK0203 Credentialed vs. non-credentialed
- AK0204 False positive

Internal Assessment Criteria

- IAC0201 Systems are scanned using appropriate tools
- IAC0202 Vulnerabilities and potential security threats to information systems are identified
- IAC0203 Consequences are identified
- IAC0204 Best way to secure IT infrastructure in an organisation is found

2.2.3 PM-02-PS03 : Evaluate the level of risk in an IT system using standard security models and apply appropriate countermeasures

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0301 Identify critical systems and critical digital assets and IT infrastructure
- PA0302 Identify types of vulnerabilities and risks in hardware and software
- PA0303 Identify the severity of possible risks, threats and vulnerabilities, impact and cost to company
- PA0304 Select and apply appropriate countermeasures

Applied Knowledge

- AK0301 Legal, policy requirements, national and international standards

Internal Assessment Criteria

- IAC0301 A sound knowledge of the possible risks and vulnerabilities that may threaten the security of an IT system, including its critical systems and digital assets, is demonstrated and countermeasures are recommended

2.2.4 PM-02-PS04 : Use appropriate tools to assess the security posture of an organisation

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0401 Use various tools appropriate to the scenario to assess the security posture of the organisation: e.g.
 - Network Security Monitoring tools
 - Encryption Tools
 - Web Vulnerability Scanning tools
 - Network Defence Wireless Tools
 - Packet Sniffers

- Antivirus Software
- Firewall
- PKI Services
- Managed Detection Services
- Penetration Testing

Applied Knowledge

- AK0401 Legal, policy requirements, national and international standards

Internal Assessment Criteria

- IAC0401 The importance of determining an organisation's security posture as well as the ways to use various appropriate tools for such assessment, is understood

2.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 industry recognised qualification with 1 years' experience in the IT industry
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

2.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

3. 252901-001-00-PM-03, Implement Protection, Prevention and Detection Measures to Mitigate Risk, Violations and Vulnerabilities, NQF Level 5, Credits 20

3.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to protect all company data, particularly sensitive data, from both internal and external threats by maintaining cybersecurity attack mitigation and incident response capability

The learner will be required to:

- PM-03-PS01 : Protect all company data, particularly sensitive data, from both internal and external threats
- PM-03-PS02 : Provide security monitoring in alignment with cybersecurity's mission to protect digital assets to a level of confidentiality, integrity and availability equal with the threat to those assets and their value to the organisation (Prevent, monitor, maintain)
- PM-03-PS03 : Detect warning signs which may indicate security breaches
- PM-03-PS04 : Assist with incidence response in case of a cybersecurity violations
- PM-03-PS05 : Understand and comply with the disaster recovery plan
- PM-03-PS06 : Effectively respond to security incidents by detecting and identification
- PM-03-PS07 : Act immediately to mitigate the impact of an incident
- PM-03-PS08 : Execute measures to remediate cybersecurity incidents
- PM-03-PS09 : Execute measures to recover from cybersecurity incidents
- PM-03-PS10 : Assess and learn lessons from the event ensuring continuous improvement

3.2 Guidelines for Practical Skills

- 3.2.1 PM-03-PS01 : Protect all company data, particularly sensitive data, from both internal and external threats**

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0101 : Design efficient security solutions to ensure common security goals
- PA0102 : Safeguard company data, particularly sensitive data, from both internal and external threats by designing broad defences against would-be intruders for a department (not multi-departmental security)
- PA0103 : Safeguard information system assets by identifying and solving potential and actual security problems
- PA0104 : Implement anti-virus systems, firewalls, data centres and software updates with a security-first mindset
- PA0105 : Protect system by defining access privileges and granting credentials to authorized users, monitor access-related activities and check for unregistered information changes
- PA0106 : Address cybersecurity controls for each critical digital asset
- PA0107 : Contribute to the development and improvement of security monitoring and incident response processes and solutions as required to support the cybersecurity program for protecting the confidentiality, integrity, and availability of information
- PA0108 : Implement security **improvements** by assessing current situation; evaluating trends; anticipating requirements
- PA0109 : Upgrade system by implementing and maintaining security controls
- PA0110 : Seek to build in security during the development stages of software systems, networks and data centres

Applied Knowledge

- AK0101 Legal, policy requirements, national and international standards
- AK0102 Types of security breaches and loss of and damage to assets
- AK0103 Types of protection measures

Internal Assessment Criteria

- IAC0101 A comprehensive knowledge of the various ways in which to protect a company's hardware, software and information and communication systems from possible breaches and would-be intruders, is demonstrated
- IAC0102 Ways in which company security may be developed, improved and upgraded are understood
- IAC0103 System is protected by defining access privileges and granting credentials to authorised users

3.2.2 PM-03-PS02 : Provide security monitoring in alignment with cybersecurity's mission to protect digital assets to a level of confidentiality, integrity and availability equal with the threat to those assets and their value to the organisation (Prevent, monitor, maintain)

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit and scenario, the learner must be able to:

- PA0201 Set up patch management systems to update applications automatically
- PA0202 Operate security monitoring and incident response toolsets with a focus on continuous improvement
- PA0203 Maintain cybersecurity attack mitigation and incident response capability for local network layer
- PA0204 Conduct vulnerability testing to identify weaknesses and collaborate with cybersecurity team to update defensive protocols as necessary
- PA0205 Extract regular reports and perform regular checks to ensure security practices are compliant
- PA0206 Check that back-ups of data and information are done or not and report - if not, it raises company vulnerability
- PA0207 Monitor network and application performance to identify an irregular activity

Applied Knowledge

- AK0201 Legal, policy requirements, national and international standards
- AK0202 Types of security breaches and loss of and damage to assets
- AK0203 Types of protection measures
- AK0204 Monitoring tools

Internal Assessment Criteria

- IAC0201 Security monitoring is provided in alignment with cybersecurity's mission to protect all digital assets to the highest level
- IAC0202 Security is built in at the development stages of software systems, networks and data centres and vulnerability tests are done throughout these initial stages
- IAC0203 Regular checks and reports are done to ensure that security practices are compliant
- IAC0204 Data back-ups are ensured and incident response toolsets are operated

3.2.3 PM-03-PS03 : Detect warning signs which may indicate security breaches

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0301 Research what popular cyber-attack methods are and how to detect them
- PA0302 Stay informed with cybersecurity trends through active involvement in forums, networking, and through popular cybercrime news outlets
- PA0303 Spot the symptoms of a data breach
- PA0304 Deploy detection technologies that make use of machine learning or AI to learn user behaviour and detect and react to anomalies
- PA0305 Know the normal system conditions and performance and identify which is NOT normal and any unusual activities or events
- PA0306 Determine security violations and inefficiencies by conducting regular checks
- PA0307 Constantly monitor for attack and run appropriate defensive protocols if breaches occur
- PA0308 Work with HR and/or team leads to educate employees on how to identify suspicious activity

Applied Knowledge

- AK0301 Types of security breaches and loss of and damage to assets
- AK0302 Types of protection measures
- AK0303 Warning signs

Internal Assessment Criteria

- IAC0301 Cyber-attack methods and cybersecurity trends are researched through active involvement in forums, networking, and through popular cybercrime news outlets
- IAC0302 The importance of a comprehensive knowledge of normal system conditions is understood, in order to detect any suspicious or abnormal activity
- IAC0303 Appropriate defensive protocols are run if breaches are detected

3.2.4 PM-03-PS04 : Assist with incidence response in case of a cybersecurity violations

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0401 Develop a cybersecurity incident response plan to contain and recover from any potential breach
- PA0402 Execute incident response procedures required by a supervisor/ line management
- PA0403 Perform mitigation activities to prevent expansion of an event (quarantine and contain) and to resolve the incident
 - When finding a potential threat or attempted breach, close off the security vulnerability
 - Isolate the damage to prevent spreading and to protect against additional vulnerabilities
 - Quarantine the corrupted servers, devices and systems so that they can be examined and made functional again
- PA0404 Participate in a coordinated response to complex/major cyber-attacks that threaten company's assets, intellectual property, and computer systems:
 - Play a role in doing the analysis, to find out how the incident occurred
 - Response functions – is there a response process in place? Develop the incidence response process, manage the communication between different teams, ensure that proper communication processes are in place and analyse the logs to see what actually happened
 - Ways to mitigate incident as it happened

- PA0405 Collect data on current security measures for risk analysis and write regular system-status reports

Applied Knowledge

- AK0401 Legal, policy requirements, national and international standards
- AK0402 Types of security breaches and loss of and damage to assets
- AK0403 Types of protection measures
- AK0404 Response plan and procedures

Internal Assessment Criteria

- IAC0401 A cybersecurity incident response plan and procedures are developed to ensure timeous intervention
- IAC0402 Security vulnerabilities are contained and quarantined until the problems/breaches are resolved
- IAC0403 The ability to participate in analysis and identification of network intrusions is demonstrated
- IAC0404 The ability to collect data on current security measures and to write regular system-status reports, is demonstrated

3.2.5 PM-03-PS05 : Understand and comply with the disaster recovery plan

Scope of Practical Skill

Given company policies and recovery plan, the learner must be able to:

- PA0501 Understand and execute a disaster recovery plan to recover from any breach and to provide business continuity after disruption from man-made or natural causes
- PA0502 Work with IT operations to set up a shared disaster recovery/business continuity plan
- PA0503 Evaluate a data recovery plan and continuously update and improve

Applied Knowledge

- AK0501 Legal, policy requirements, national and international standards
- AK0502 Types of security breaches and loss of and damage to assets
- AK0503 Types of protection measures
- AK0504 Response plan and procedures

Internal Assessment Criteria

- IAC0501 Requirements of a disaster recovery plan is understood and applied
- IAC0502 A data recovery plan is developed and continuously updated and improved upon

3.2.6 PM-03-PS06 : Effectively respond to security incidents by detecting and identification

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0601 Monitor the network and systems so that any irregularities are flagged immediately.
- PA0602 Detect and ascertain the source of the cybersecurity incident
- PA0603 Determine the nature of the incident or combination of threats
- PA0604 Identify what aspects of your system have been compromised
- PA0605 Assess potential damage and severity
- PA0606 Determine the impact, what information was lost/stolen, whether any client or critical information was compromised or standards and contracts were violated.
- PA0607 Begin the notification process

- PA0608 Collect evidence each step along the way e.g. take screen shots, paste into word document and save on central shared folder.

Applied Knowledge

- AK0601 Legal, policy requirements, national and international standards
- AK0602 Types of security breaches and loss of and damage to assets
- AK0603 Types of protection measures
- AK0604 Response plan and procedures

Internal Assessment Criteria

- IAC0601 Apply the correct incidence response plans and steps to mitigate the risk and recorded

3.2.7 PM-03-PS07 : Act immediately to mitigate the impact of an incident

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0701 Execute immediate steps to contain the breach in the short term, e.g. shutting down programs or systems and disconnecting from networks
- PA0702 Identify and quarantine any malware discovered
- PA0703 Execute measures to contain the breach and prevent future breaches in the long term e.g. update protections as needed, review and strengthen access credentials as necessary
- PA0704 Identify and remove any personnel involved
- PA0705 Collect evidence each step along the way e.g. take screen shots, paste into word document and save on central shared folder.

Applied Knowledge

- AK0701 Legal, policy requirements, national and international standards
- AK0702 Types of security breaches and loss of and damage to assets
- AK0703 Types of protection measures
- AK0704 Response plan and procedures

Internal Assessment Criteria

- IAC0701 Immediate actions are executed to contain the damage and recorded

3.2.8 PM-03-PS08 : Execute measures to remediate cybersecurity incidents

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0801 Ensure all artefacts of the incident have been fully removed from the system
- PA0802 Repair or update systems as needed
- PA0803 Check that all software patches are current and protections strengthened
- PA0804 Ensure back-ups are in place and functioning properly
- PA0805 Collect evidence each step along the way e.g. take screen shots, paste into word document and save on central shared folder.

Applied Knowledge

- AK0801 Remediation steps and actions
- AK0802 Legal, policy requirements, national and international standards

- AK0803 Types of security breaches and loss of and damage to assets
- AK0804 Types of protection measures
- AK0805 Response plan and procedures

Internal Assessment Criteria

- IAC0801 The remediation process is successfully concluded and documented.

3.2.9 PM-03-PS09 : Execute measures to recover from cybersecurity incidents

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA0901 Reinstall the environment.
- PA0902 Restore and set up the computer from scratch, commission, reconnect and complete configuration
- PA0903 Continuously monitor to ensure the incident was fully resolved and no further threats are detected.
- PA0904 Recover data, restore back-ups and check for proper functionality
- PA0905 Record and document procedures, steps and findings
- PA0906 Identify the impact on insurance requirements
- PA0907 Communicate with internal and external stakeholders and inform them of the breach, if there is a need to do so
- PA0908 Provide information in the form of a wire diagram
- PA0909 Collect evidence each step along the way e.g. take screen shots, paste into word document and save on central shared folder.

Applied Knowledge

- AK0901 Legal, policy requirements, national and international standards
- AK0902 Types of security breaches and loss of and damage to assets
- AK0903 Types of protection measures
- AK0904 Response plan and procedures

Internal Assessment Criteria

- IAC0901 After back-ups have been restored and impact of lost/stolen data has been determined, computer network is set up from scratch and reconnected and configuration is completed
- IAC0902 All stake-holders are informed of any breach/theft and are given an assurance as soon as security issues are resolved
- IAC0903 A knowledge of insurance requirements is displayed

3.2.10 PM-03-PS10 : Assess and learn lessons from the event ensuring continuous improvement

Scope of Practical Skill

Given access to a cybersecurity toolkit/platform, a virtual environment suitable to the toolkit, and scenario, the learner must be able to:

- PA1001 Perform assessment on the breach, document the findings, compile a report (close-out report), and submit to the community
- PA1002 Report on and present information according to generally accepted standards.

- PA1003 Implement improvements by incorporating lessons learned from current and previous detection / response activities
- PA1004 Replace old security technology with new and create new defence including solid processes and controls in both IT and the business
- PA1005 Research for new standard being introduced into the market and make sure the system is compliant

Applied Knowledge

- AK1001 Legal, policy requirements, national and international standards
- AK1002 Types of security breaches and loss of and damage to assets
- AK1003 Types of protection measures
- AK1004 Response plan and procedures

Internal Assessment Criteria

- IAC1001 Security is ensured by installing the newest security technology

3.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 industry recognised qualification with 1 years' experience in the IT industry
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

3.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

4. 252901-001-00-PM-04, Apply Logical Thinking and Maths, NQF Level 4, Credits 6

4.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to use basic maths as a basis for understanding maths encountered in the programming within the field of networking

The learner will be required to:

- PM-04-PS01 : Number bases and measurement units
- PM-04-PS02 : Basic math
- PM-04-PS03 : Operator precedence
- PM-04-PS04 : Integer division
- PM-04-PS05 : Modulus
- PM-04-PS06 : Increments

4.2 Guidelines for Practical Skills

4.2.1 PM-04-PS01 : Number bases and measurement units

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0101 Convert numbers between the decimal number system and the binary number system: addition and subtraction of positive whole numbers in binary up to 100002 (16 in decimal, binary and hex)
- PA0102 Express size and magnitude: From Giga to Pica (10^{12} to 10^{-12})
- PA0103 Apply conversions between related units in different measurement systems
- PA0104 Apply rational and irrational numbers
- PA0105 Differentiate and apply PEDMAS and BODMAS
- PA0106 Use decimals and convert them to common fraction form
- PA0107 Apply scientific notation for small and large numbers.

Applied Knowledge

- AK0101 The effect of error in calculations

Internal Assessment Criteria

- IAC0101 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated

4.2.2 PM-04-PS02 : Basic math

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0201 Arithmetic expressions in programming: Addition (+), subtraction (-), multiplication (*) and division (/) for both integers and real data types
- PA0202 Percentage
- PA0203 Negative numbers
- PA0204 Cartesian coordinate system
- PA0205 Pythagorean theorem for finding the distance between two points
- PA0206 Decimal, binary, and hexadecimal numbering systems

Applied Knowledge

- AK0201 The effect of error in calculations

Internal Assessment Criteria

- IAC0201 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated

4.2.3 PM-04-PS03 : Operator precedence

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0301 Use parentheses
- PA0302 Apply order of operations (left to right)
- PA0303 Apply precedence:
 - Parentheses
 - Multiply and Divide
 - Add and Subtract
- PA0304 Use PEDMAS

Applied Knowledge

- AK0301 The effect of error in calculations

Internal Assessment Criteria

- IAC0301 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated

4.2.4 PM-04-PS04 : Integer division

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0401 Divide two integers
- PA0402 Discard the fractional part (remainder)

Applied Knowledge

- AK0401 The effect of error in calculations

Internal Assessment Criteria

- IAC0401 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated.

4.2.5 PM-04-PS05 : Modulus

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0501 Modulus abbreviated as “mod”
- PA0502 %, also known as the modulus or remainder operator
- PA0503 The % operator returns the remainder of the two numbers after division
 - what is left after what is in front of the modulus operator is divided by what is after the modulus operator
 - Examples
- PA0504 Use % to check if a number is even or odd
- PA0505 The double equals (==) operator is used to compare values, such as a pair of ints, and returns a Boolean

Applied Knowledge

- AK0501 The effect of error in calculations

Internal Assessment Criteria

- IAC0501 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated

4.2.6 PM-04-PS06 : Increments

Scope of Practical Skill

Given a demonstration, thereafter suitable problem statements and tasks, the learner must be able to:

- PA0601 Increment a variable by 1 using $n = n + 1$
- PA0602 $n = n + 1$ means n takes the new value of whatever $n + 1$ is
- PA0603 compound assignment operator using $n += 1$ or $n++$
- PA0604 Add one to n : $n + 1$
- PA0605 Add two to n : $n = n + 2$ or $n += 2$
- PA0606 $++$ operator can only add 1

Applied Knowledge

- AK0601 The effect of error in calculations

Internal Assessment Criteria

- IAC0601 An ability to apply mathematical thinking to solve a mathematical problem is demonstrated

4.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 industry recognised qualification with 1 years' relevant experience
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate

- Ethical clearance (where necessary)

4.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

5. 252901-001-00-PM-05, Apply Basic Scriptwriting for Cybersecurity Toolsets, NQF Level 4, Credits 4

5.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to apply basic scripting skills to use a toolsets in the field of study or employment

The learner will be required to:

- PM-05-PS01 : Source and compare at least three software toolkits/platforms/ languages used in your field of studies
- PM-05-PS02 : Set up an editing environment (tailored to a specific tool or platform)
- PM-05-PS03 : Write a script using a Command Line Interface/Terminal sessions for giving instructions for use of a toolset
- PM-05-PS04 : Script write loops (tailored to a specific tool or platform)
- PM-05-PS05 : Handle errors (tailored to a specific tool or platform)
- PM-05-PS06 : Apply general steps for writing script (tailored to a specific tool or platform)
- PM-05-PS07 : Practical exercise using the specified product set

5.2 Guidelines for Practical Skills

5.2.1 PM-05-PS01 : Source and compare at least three software toolkits/platforms/ languages used in your field of studies

Scope of Practical Skill

Given access to the internet, the learner must be able to:

- PA0101 Identify at least three most suitable software toolkits/platforms in the field of studies or employment
- PA0102 Source these from the internet
- PA0103 Compare these in terms of advantages and disadvantages related to the field of study or employment

Applied Knowledge

- AK0101 Software toolkit / software
- AK0102 Computer hardware

Internal Assessment Criteria

- IAC0101 At least three suitable software toolkits/platforms/languages necessary for field of study are sourced from the internet and compared for suitability in terms of advantages/disadvantages

5.2.2 PM-05-PS02 : Set up an editing environment (tailored to a specific tool or platform)

Scope of Practical Skill

Given a problem statement and access to a suitable PC or device with software toolkit/platform/, the learner must be able to:

- PA0201 Choose an editor or utility in line with the application and the paradigm
- PA0202 Install a programming environment and/or language specific to a programming language and configure for use

Applied Knowledge

- AK0201 Cybersecurity utility
- AK0202 Software toolkit / software
- AK0203 Hardware

Internal Assessment Criteria

- IAC0201 A programming environment is created and a programming language that is in line with the chosen paradigm (e.g., Java) is configured for use in the environment

5.2.3 PM-05-PS03 : Write a script using a Command Line Interface/Terminal sessions for giving instructions for use of a toolset

Scope of Practical Skill

Given a problem statement and access to a suitable PC or device with security software toolset, the learner must be able to:

- PA0301 Write a script using a Command Line Interface/Terminal sessions for giving instructions for use of a toolset
- PA0302 Select and use correct data types (tailored to a specific tool or platform)
- PA0303 Use complex types to organise data and create expressions
- PA0304 Use and apply: If-else statements, Booleans
- PA0305 SQL injection attacks: Inject special characters

Applied Knowledge

- AK0301 Coding language
- AK0302 Software toolkit / software
- AK0303 Hardware

Internal Assessment Criteria

- IAC0301 Script is written using a CLI
- IAC0302 Correct data types are identified as pertaining to the scripting activity
- IAC0303 Principles of complex types are applied during scripting

5.2.4 PM-05-PS04 : Script write loops (tailored to a specific tool or platform)

Scope of Practical Skill

Given a problem statement and access to a suitable PC or device with software toolkit/platform, the learner must be able to:

- PA0401 Use and apply:
 - While loops
 - Iterating loops
 - Nested loops

Applied Knowledge

- AK0401 Types of loops
- AK0402 Checking conditions
- AK0403 Initializer, condition, iterator
- AK0404 Sequence

Internal Assessment Criteria

- IAC0401 Use loops to repeat operations

5.2.5 PM-05-PS05 : Handle errors (tailored to a specific tool or platform)

Scope of Practical Skill

Given a problem statement and access to a suitable PC or device with software toolkit/platform, the learner must be able to:

- PA0501 Identify errors peculiar to the platform and how best to handle them
 - Bugs, testing and debugging
 - Exceptions: thrown and caught and logged
 - Exception handling
 - Code the application to recover by itself
 - Error tracking
 - Fix errors

Applied Knowledge

- AK0501 Types of errors
- AK0502 Types of testing
- AK0503 Possible causes
- AK0504 Anticipation, detection, and resolution of programming, application, and communications errors

Internal Assessment Criteria

- IAC0501 Handle various types of errors

5.2.6 PM-05-PS06 : Apply general steps for writing script (tailored to a specific tool or platform)

Scope of Practical Skill

Given a problem statement and access to a suitable PC or device with software toolkit/platform, the learner must be able to:

- PA0601 Understand the problem you are trying to solve
- PA0602 Design a solution
- PA0603 Draw a flow chart
- PA0604 Write pseudo-code
- PA0605 Write script
- PA0606 Test and debug
- PA0607 Insert comments in script and provide proper documentation that will guide other developers through the algorithm and logic that was implemented.
- PA0608 Finalise documentation
- PA0609 Test with real-world users
- PA0610 Release script
- PA0611 Iterate the steps for next version

Applied Knowledge

- AK0601 Command line interface (CLI) and minimum commands
- AK0602 Using cybersecurity software toolset

Internal Assessment Criteria

- IAC0601 The general steps for writing a script are understood and applied and results tested before the program is released and the steps repeated for the next version

5.2.7 PM-05-PS07 : Practical exercise using the specified product set

Scope of Practical Skill

Given access to a suitable PC or device with software toolkit/platform the learner must be able to:

- PA0701 Create a text file
- PA0702 Write a script that extract MAC (media access control)
- PA0703 Pull up current user name and drop into text file
- PA0704 Write script that extracts information on the current security posture of a machine within its environment

Applied Knowledge

- AK0701 Language
- AK0702 Coding
- AK0703 Using software toolkit / platform

Internal Assessment Criteria

- IAC0701 Desired outcome is achieved

5.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 industry recognised qualification with 1 years' experience in the IT industry
 - CYBERSECURITY vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

5.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

6. 252901-001-00-PM-06, Access and Visualise Structured Data Using Spreadsheets, NQF Level 4, Credits 5

6.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to analyse and visualise data

The learner will be required to:

- PM-06-PS01 : Report data
- PM-06-PS02 : Summarise and format data using tables
- PM-06-PS03 : Create, use and edit pivot tables and pivot charts
- PM-06-PS04 : Create, use and edit dashboards
- PM-06-PS05 : Create and configure hierarchies and time data
- PM-06-PS06 : Apply a data model
- PM-06-PS07 : Import data from files
- PM-06-PS08 : Import data from databases
- PM-06-PS09 : Import data from reports
- PM-06-PS10 : Visualize data

6.2 Guidelines for Practical Skills

6.2.1 PM-06-PS01 : Report data

Scope of Practical Skill

Given a suitable problem to solve, the learner must be able to:

- PA0101 Create report
- PA0102 Filter and format data
- PA0103 Create charts

Applied Knowledge

- AK0101 Data analysis
- AK0102 Functions and capabilities

Internal Assessment Criteria

- IAC0101 Reporting is done by creating charts containing the correct data

6.2.2 PM-06-PS02 : Summarise and format data using tables

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0201 Create a table
- PA0202 Summarise data
- PA0203 Sort, filter, and validate data
- PA0204 Format summarized data

Applied Knowledge

- AK0201 Data analysis
- AK0202 Functions and capabilities

Internal Assessment Criteria

- IAC0201 A table is created and summarized data sorted, filtered, validated and formatted

6.2.3 PM-06-PS03 : Create, use and edit pivot tables and pivot charts

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0301 Use pivot tables and pivot charts
- PA0302 Import data from a CSV file
- PA0303 Create a pivot table
- PA0304 Edit pivot tables and pivot charts

Applied Knowledge

- AK0301 Data analysis
- AK0302 Functions and capabilities

Internal Assessment Criteria

- IAC0301 Pivot tables and charts are created, used and edited, using data imported from a CSV file

6.2.4 PM-06-PS04 : Create, use and edit dashboards

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0401 Create dashboards
- PA0402 Conduct data analysis in excel pivot tables
- PA0403 Arrange tables and charts
- PA0404 Slice data
- PA0405 Filter data using a slicer
- PA0406 Add calculated columns to a dashboard
- PA0407 Find anomalies

Applied Knowledge

- AK0401 Data analysis
- AK0402 Functions and capabilities

Internal Assessment Criteria

- IAC0401 Dashboards are created containing data backed tables and charts, data are sliced, filtered and calculated and anomalies found

6.2.5 PM-06-PS05 : Create and configure hierarchies and time data

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0501 Create a hierarchy
- PA0502 Configure time data
- PA0503 Create an animated time chart

Applied Knowledge

- AK0501 Data analysis

- AK0502 Functions and capabilities

Internal Assessment Criteria

- IAC0501 Hierarchies are created, time data is configured and an animated time chart is created

6.2.6 PM-06-PS06 : Apply a data model

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0601 Explore an data model
- PA0602 Add multiple tables
- PA0603 Create relationships
- PA0604 Add external data
- PA0605 Import external data and use it
- PA0606 Link out to external data
- PA0607 Use the DAX (data analysis expressions) function
- PA0608 View data within a data table

Applied Knowledge

- AK0601 Data analysis
- AK0602 Functions and capabilities

Internal Assessment Criteria

- IAC0601 A data model is applied for integrating data from various imported tables and creating relationships between them

6.2.7 PM-06-PS07 : Import data from files

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0701 Pre-format and import CSV files
- PA0702 Import data into the software program
- PA0703 Shape and transform data
- PA0704 Load data

Applied Knowledge

- AK0701 Data analysis
- AK0702 Functions and capabilities

Internal Assessment Criteria

- IAC0701 Data are imported from pre-formatted files into excel and configured

6.2.8 PM-06-PS08 : Import data from databases

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0801 Import data from a SQL server database
- PA0802 Identify available data sources

- PA0803 Preview, shape, and transform data
- PA0804 Table relationships and hierarchies
- PA0805 Load data

Applied Knowledge

- AK0801 Data analysis
- AK0802 Functions and capabilities

Internal Assessment Criteria

- IAC0801 Data are imported from various identified databases and configured, taking table relationships and hierarchies into account

6.2.9 PM-06-PS09 : Import data from reports

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA0901 Import data from reports
- PA0902 Transform report data

Applied Knowledge

- AK0901 Data analysis
- AK0902 Functions and capabilities

Internal Assessment Criteria

- IAC0901 Data are imported from s reports and converted

6.2.10 PM-06-PS10 : Visualize data

Scope of Practical Skill

Given a suitable problem to solve, a PC or laptop and software, the learner must be able to:

- PA1001 Create and format measures
- PA1002 Visualize data using:
 - Pivot charts
 - Cube functions
 - Charts for cube functions
- PA1003 Create and refine a pivot chart

Applied Knowledge

- AK1001 Cube functions and when to use them
- AK1002 A number of charts for use with cube functions

Internal Assessment Criteria

- IAC1001 Data is visualized in excel with created and refined pivot charts and charts with cube functions in order to produce a graphical representation of a set of data

6.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.

- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 6 industry recognised qualification with 1 years' experience in the IT industry
 - Cybersecurity vendor certification
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

6.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

7. 252901-001-00-PM-07, Applying Design Thinking Methodologies, NQF Level 4, Credits 4

7.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to participate in a design thinking intervention and apply design thinking methodologies and look for opportunities to apply the same methodology in world-of-work and personal life

The learner will be required to:

- PM-07-PS01 : Collaborate with team members to apply innovative and problem solving strategies.
- PM-07-PS02 : Apply design thinking process to solve a problem creatively and innovatively

7.2 Guidelines for Practical Skills

7.2.1 PM-07-PS01 : Collaborate with team members to apply innovative and problem solving strategies.

Scope of Practical Skill

Given access to a Design Thinking workshop with multidisciplinary team members, the learner must be able to:

- PA0101 Interact and collaborate with team members.
- PA0102 Engage in dialogues as required by the process.
- PA0103 Participate in the activities as presented by the facilitator.
- PA0104 Actively contribute to the discussions and activities.

Applied Knowledge

- AK0101 Design Thinking as a solution-based approach to solving problems.

Internal Assessment Criteria

- IAC0101 The ability to collaborate with team members towards a joint goal is demonstrated.

7.2.2 PM-07-PS02 : Apply design thinking process to solve a problem creatively and innovatively

Scope of Practical Skill

Given access to a Design Thinking workshop with multidisciplinary team members and techniques and equipment, the learner must be able to:

- PA0201 Empathise with users/customers by setting aside own assumptions and formulate real insights about the user
- PA0202 Define the problem statement and create a point-of-view statement, indicating a specific user, a need, and an insight
- PA0203 Ideate by challenging assumptions and creating ideas for innovative solutions to help resolve the challenges and needs
- PA0204 Prototype workable ideas
- PA0205 Test the solution applying a reiterative process and using various techniques

Applied Knowledge

- AK0201 Involving five phases: Empathize, Define, Ideate, Prototype and Test

Internal Assessment Criteria

- IAC0201 The ability to participate in the design thinking phases is demonstrated.

- IAC0202 Use skills to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test and innovative and creative solution

7.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 industry recognised qualification with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

7.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

8. 252901-001-00-PM-08, Function Ethically and Effectively as a Member of a Multidisciplinary Team, NQF Level 4, Credits 5

8.1 Purpose of the Practical Skills Module

The focus of the learning in this module is on providing the learner with an opportunity to acquire the skills to function ethically and effectively in the workplace

The learner will be required to:

- PM-08-PS01 : Present information to an audience
- PM-08-PS02 : Conduct basic research (gather and explore data and information) on 4IR skills and application opportunities in the workplace
- PM-08-PS03 : Ensure compliance with the code of conduct and governance in the workplace
- PM-08-PS04 : Collaborate with team members in the workplace
- PM-08-PS05 : Attend and participate in meetings

8.2 Guidelines for Practical Skills

8.2.1 PM-08-PS01 : Present information to an audience

Scope of Practical Skill

Given information on an audience and presentation tools, the learner must be able to:

- PA0101 Analyse the audience and research if needed
- PA0102 Select a topic
- PA0103 Define the objective and conduct research on the topic or gather information from an available source.
- PA0104 Prepare the content and body of the presentation
- PA0105 Visualise the content using suitable technology
- PA0106 Decide on the method of presentation e.g. story telling
- PA0107 Prepare the introduction and conclusion
- PA0108 Prepare for the presentation
- PA0109 Adapt presentation style to the audience and reaction of the audience.
- PA0110 Speak from memory, from notes and from text.

Applied Knowledge

- AK0101 Presentation principles
- AK0102 Presentation methods and techniques
- AK0103 Presentation technologies appropriate to the information

Internal Assessment Criteria

- IAC0101 Information is presented using tools, techniques and presentation methods appropriate to the audience.

8.2.2 PM-08-PS02 : Conduct basic research (gather and explore data and information) on 4IR skills and application opportunities in the workplace

Scope of Practical Skill

Given access to the internet, the learner must be able to:

- PA0201 Gather and explore information on technologies such as Google, Amazon and MS and identify opportunities to use these tools to improve or reduce development time (e.g. embed AI APIs).
- PA0202 Gather and explore information on business intelligence (BI) applications and availability of Big Data (collecting data, converting data into information and turning information into knowledge, knowledge into intelligence and intelligence into wisdom)

Applied Knowledge

- AK0201 4IR technologies
- AK0202 Impact of 4IR on businesses and individuals.

Internal Assessment Criteria

- IAC0201 The application of future skills and competencies in the workplace is researched and opportunities to adapt to the requirements of 4IR are identified.

8.2.3 PM-08-PS03 : Ensure compliance with the code of conduct and governance in the workplace

Scope of Practical Skill

Given legislation, industry norms and standards, company policy on governance and code of conduct, the learner must be able to:

- PA0301 Compare company policy with legislation, industry norms and standards to identify any shortfalls or opportunities for improvement
- PA0302 Compare code of conduct with the policy and identify any shortfalls or opportunities for improvement
- PA0303 Compare IT procedures with policy and code of conduct and identify any shortfalls or opportunities for improvement

Applied Knowledge

- AK0301 Ethics: concept and principles
- AK0302 Function of governance
- AK0303 Compliance vs non-compliance

Internal Assessment Criteria

- IAC0301 Company governance policy and procedures are assessed for compliance with legislation and industry norms and standards.

8.2.4 PM-08-PS04 : Collaborate with team members in the workplace

Scope of Practical Skill

Given access to a communication tool (e.g. Agile) and team members, the learner must be able to:

- PA0401 Use technology to communicate and report on progress
- PA0402 Actively collaborate with team members to achieve shared goals
- PA0403 Resolve conflict (if it occurs) using conflict resolution and other interpersonal skills
- PA0404 Apply time management skills and problem solving skills
- PA0405 Achieve goals timeously in support of team goals.

Applied Knowledge

- AK0401 Reporting and communication technology

Internal Assessment Criteria

- IAC0401 Management and team members are informed about progress through communication and reporting.
- IAC0402 Interpersonal and intrapersonal skills are applied to achieve own and team goals.

8.2.5 PM-08-PS05 : Attend and participate in meetings

Scope of Practical Skill

Given role play on meetings, the learner must be able to:

- PA0501 Apply meeting procedures
- PA0502 Prepare the agenda for the meeting
- PA0503 Take notes during the meeting and produce minutes of the meeting.
- PA0504 Participate and contribute to the discussions in the meeting.

Applied Knowledge

- AK0501 Meeting procedures

Internal Assessment Criteria

- IAC0501 Meeting procedures are observed.
- IAC0502 Documentation (agenda and minutes) are produced which give an accurate representation of the agreements.

8.3 Provider Programme Accreditation Criteria

Physical Requirements:

- Valid licenses software and application, including OS.
- Internet connection and hardware availability
- Examples and information specified in the scope statement and all the case studies, scenarios and access to hardware and software implied in the scope statements of the modules.
- Remote learners: Provider must provide business IT simulation system (e.g. invoice processing).

Human Resource Requirements:

- Lecturer/learner ratio of 1:10 (Maximum)
- Qualification of lecturer (SME):
 - NQF 5 industry recognised qualification with 1 years' relevant experience
- Assessors and moderators: accredited by the MICT SETA

Legal Requirements:

- Legal (product) licences to use the software for learning and training
- OHS compliance certificate
- Ethical clearance (where necessary)

8.4 Exemptions

- No exemptions, but the module can be achieved in full through a normal RPL process

SECTION 3C: WORK EXPERIENCE MODULE SPECIFICATIONS

List of Work Experience Module Specifications

Work Experience	252901-001-00-WM-01	Compliance with Legal Cybersecurity Requirements	5	12
Work Experience	252901-001-00-WM-02	Cybersecurity Risk Assessment and Mitigation	5	20
Work Experience	252901-001-00-WM-03	Cybersecurity Detection, Protection and Prevention Processes	5	20

1. 252901-001-00-WM-01, Compliance with Legal Cybersecurity Requirements, NQF Level 5, Credits 12

1.1 Purpose of the Work Experience Module

The focus of the work experience is on providing the learner with an opportunity to:

Demonstrate knowledge and understanding of cybersecurity concepts and investigate how cybersecurity affects legal compliance and solidarity in companies and communities

The learner will be required to:

- WM-01-WE01 : Attend induction program and familiarise self with company processes, procedures, tools and culture
- WM-01-WE02 : Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.
- WM-01-WE03 : Conduct the following tasks under supervision.

1.2 Guidelines for Work Experience

1.2.1 WM-01-WE01 : Attend induction program and familiarise self with company processes, procedures, tools and culture

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0101 Attend induction program and familiarise self with the culture of the company.
- WA0102 Apply protocols and work etiquette.
- WA0103 Attend company specific information sharing sessions (e.g. standing meetings, toolbox talks, power hours, etc.)
- WA0104 Familiarise self with and apply “working from anywhere” protocols.
- WA0105 Read and understand company cybersecurity policy, protocols and procedures.
- WA0106 Comply with governance protocols and code of ethics of the company and ensure legal compliance by adhering to legal requirements (incl. but not limited to privacy, confidentiality, security of data, etc.).
- WA0107 Spend time in the various departments of the company, observe process flows and compile wire diagrams or workflow of the processes observed using suitable tools and showing the relationships and influences each of the departments have on each other.
- WA0108 Understand management requirements and expectations from cybersecurity measures.
- WA0109 Understand cybersecurity protocols and procedures.
- WA0110 Understand company assets in terms of cybersecurity.
- WA0111 Manage timesheets and apply self-management skills.
- WA0112 Collaborate with team members to achieve common and individual goals

Supporting Evidence

- SE0101 Attendance registers
- SE0102 Wire diagrams or workflows

1.2.2 WM-01-WE02 : Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0201 Comply with requirements of risk management frameworks
- WA0202 Comply with security regulations and standards
- WA0203 Monitor compliance with information security policies and procedures
- WA0204 Give advice and guidance to staff on issues such as spam and unwanted or malicious emails.
- WA0205 Train organisation on security measures
- WA0206 Maintain an information security risk register and assist with internal and external audits relating to information security
- WA0207 Administering information security software and controls
- WA0208 Maintaining security records of monitoring and incident response activities
- WA0209 Document security breaches and assess the damage they cause.
- WA0210 Producing situational and incident-related reports
- WA0211 Providing timely and relevant security reports
- WA0212 Generate reports for both technical and non-technical staff and stakeholders

Supporting Evidence

- SE0201 Logs and security reports

1.2.3 WM-01-WE03 : Conduct the following tasks under supervision.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0301 Comply with requirements of risk management frameworks
- WA0302 Comply with security regulations and standards
- WA0303 Monitor compliance with information security policies and procedures
- WA0304 Give advice and guidance to staff on issues such as spam and unwanted or malicious emails.
- WA0305 Train organisation on security measures
- WA0306 Maintain an information security risk register and assist with internal and external audits relating to information security
- WA0307 Administering information security software and controls
- WA0308 Maintaining security records of monitoring and incident response activities
- WA0309 Document security breaches and assess the damage they cause.
- WA0310 Producing situational and incident-related reports
- WA0311 Providing timely and relevant security reports
- WA0312 Generate reports for both technical and non-technical staff and stakeholders

Supporting Evidence

- SE0301 Logs and security reports

1.3 Contextualised Workplace Knowledge

1. Organisational policies, procedures and guidelines which relate to cybersecurity.
2. Standard templates, documentation and tools.
3. Business processes and workflows pertaining to cybersecurity.
4. Team members.
5. Company specific operating systems.
6. Vendor specific cybersecurity tools and platforms.
7. Company specific best practices, quality standards and security protocols.

8. Version control system.
9. Sources of information available for testing.
10. Approval process

1.4 Criteria for Workplace Approval

Physical Requirements:

- Tools, equipment, systems, e.g.: company systems, documents, data, relevant meetings, teams and supervisors, design studio, etc.
- Key processes, e.g.: CYBERSECURITY design, testing and deployment processes project on the go or completed.

Human Resource Requirements:

- Maximum mentor/learner ratio of 1:3 in the ideal situation
- Supervisor/mentor: 2 years' software development experience

Legal Requirements:

- Legal (product) licences to use software
- OHS compliance certificate
- Ethical clearance (where necessary)

1.5 Additional Assignments to be Assessed Externally

None

2. 252901-001-00-WM-02, Cybersecurity Risk Assessment and Mitigation, NQF Level 5, Credits 20

2.1 Purpose of the Work Experience Module

The focus of the work experience is on providing the learner with an opportunity to:

Assess risk to assets and evaluate current cybersecurity protection measures

The learner will be required to:

- WM-02-WE01 : Shadow and observe an experienced Cybersecurity Analyst undertaking a cybersecurity risk assessment.
- WM-02-WE02 : Conduct a cybersecurity risk assessment under supervision.

2.2 Guidelines for Work Experience

2.2.1 WM-02-WE01 : Shadow and observe an experienced Cybersecurity Analyst undertaking a cybersecurity risk assessment.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0101 Determine information value
- WA0102 Identify and prioritise assets
- WA0103 Identify cyber threats
 - Performing cyber and technical threat analyses
 - Research/evaluate emerging cyber security threats and ways to manage them
 - Evaluate and deconstruct malware software
 - Stay current on it security trends and news.
 - Stay up-to-date on information technology trends and security standards.
- WA0104 Identify vulnerabilities
 - Discovering vulnerabilities in information systems
 - Identify potential weaknesses and implement measures, such as firewalls and encryption
 - Performing tests and risk analysis
- WA0105 Analyse controls and the need for new controls
 - Stay one step ahead of cyber attacks
- WA0106 Calculate the likelihood and impact of various scenarios on a per-year basis
- WA0107 Prioritize risks based on the cost of prevention vs information value
- WA0108 Document results in risk assessment report

Supporting Evidence

- SE0101 Attendance registers and risk assessment report

2.2.2 WM-02-WE02 : Conduct a cybersecurity risk assessment under supervision.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0201 Determine information value
- WA0202 Identify and prioritise assets
- WA0203 Identify cyber threats

- Performing cyber and technical threat analyses
- Research/evaluate emerging cyber security threats and ways to manage them
- Evaluate and deconstruct malware software
- Stay current on it security trends and news.
- Stay up-to-date on information technology trends and security standards.
- WA0204 Identify vulnerabilities
 - Discovering vulnerabilities in information systems
 - Identify potential weaknesses and implement measures, such as firewalls and encryption
 - Performing tests and risk analysis
- WA0205 Analyse controls and the need for new controls
 - Stay one step ahead of cyber attacks
- WA0206 Calculate the likelihood and impact of various scenarios on a per-year basis
- WA0207 Prioritize risks based on the cost of prevention vs information value
- WA0208 Document results in risk assessment report

Supporting Evidence

- SE0201 Attendance register
- SE0202 Report on findings in suitable format

2.3 Contextualised Workplace Knowledge

1. Organisational policies, procedures and guidelines which relate to cybersecurity.
2. Standard templates, documentation and tools.
3. Business processes and workflows pertaining to cybersecurity.
4. Team members.
5. Company specific operating systems.
6. Vendor specific cybersecurity tools and platforms.
7. Company specific best practices, quality standards and security protocols.
8. Version control system.
9. Sources of information available for testing.
10. Approval process

2.4 Criteria for Workplace Approval

Physical Requirements:

- Tools, equipment, systems, e.g.: company systems, documents, data, relevant meetings, teams and supervisors, design studio, etc.
- Key processes, e.g.: CYBERSECURITY design, testing and deployment processes project on the go or completed.

Human Resource Requirements:

- Maximum mentor/learner ratio of 1:3 in the ideal situation
- Supervisor/mentor: 2 years' software development experience

Legal Requirements:

- Legal (product) licences to use software

- OHS compliance certificate
- Ethical clearance (where necessary)

2.5 Additional Assignments to be Assessed Externally

None

3. 252901-001-00-WM-03, Cybersecurity Detection, Protection and Prevention Processes, NQF Level 5, Credits 20

3.1 Purpose of the Work Experience Module

The focus of the work experience is on providing the learner with an opportunity to:

Conduct a cybersecurity risk assessment under supervision and apply detection, protection and prevention processes.

The learner will be required to:

- WM-03-WE01 : Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.
- WM-03-WE02 : Conduct a cybersecurity risk assessment under supervision.

3.2 Guidelines for Work Experience

3.2.1 WM-03-WE01 : Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0101 **Protect**
 - Implement intrusion detection systems (IDS)
 - Implement security information event management (SIEM)
 - Implement threat plans and protocols,
 - Perform penetration testing
 - Perform security monitoring
 - Upgrade existing security systems
 - Engage in 'ethical hacking', for example, simulating security breaches
- WA0102 **Prevent**
 - Apply intrusion prevention tools
 - Help colleagues install security software and understand information security management.
 - Implement and maintain security frameworks for existing and new systems
 - Install, update and maintain security measures and operate software to protect systems and information infrastructure, including firewalls and data encryption programs.
 - Improve security efficiency
- WA0103 **Detect**
 - Use analytic tools to determine emerging threat patterns and vulnerabilities
 - Investigate security alerts and provide incident response
 - Investigate security breaches and other cybersecurity incidents.
 - Monitor and respond to 'phishing' emails and 'pharming' activity
 - Monitor computer networks for security issues.
 - Monitor for attacks, intrusions and unusual, unauthorised or illegal activity
 - Monitor identity and access management, including monitoring for abuse of permissions by authorised system users
 - Monitoring security network access,
 - Analysing security system logs, security tools, and data
 - Review security alerts and taking steps to protect the information,

- **WA0104 Respond and recover**
 - Plan for disaster recovery and create contingency plans in the event of any security breaches
 - Respond to security events
 - Fix detected vulnerabilities to maintain a high-security standard.
 - Apply technical knowledge in data loss prevention

Supporting Evidence

- SE0101 Risk assessment and logs

3.2.2 WM-03-WE02 : Conduct a cybersecurity risk assessment under supervision.

Scope of Work Experience

The person will be expected to engage in the following work activities:

- **WA0201 Protect**
 - Implement intrusion detection systems (IDS)
 - Implement security information event management (SIEM)
 - Implement threat plans and protocols,
 - Perform penetration testing
 - Perform security monitoring
 - Upgrade existing security systems
 - Engage in 'ethical hacking', for example, simulating security breaches
- **WA0202 Prevent**
 - Apply intrusion prevention tools
 - Help colleagues install security software and understand information security management.
 - Implement and maintain security frameworks for existing and new systems
 - Install, update and maintain security measures and operate software to protect systems and information infrastructure, including firewalls and data encryption programs.
 - Improve security efficiency
 - Prevent hacker intrusion
- **WA0203 Detect**
 - Use analytic tools to determine emerging threat patterns and vulnerabilities
 - Investigate security alerts and provide incident response
 - Investigate security breaches and other cybersecurity incidents.
 - Monitor and respond to 'phishing' emails and 'pharming' activity
 - Monitor computer networks for security issues.
 - Monitor for attacks, intrusions and unusual, unauthorised or illegal activity
 - Monitor identity and access management, including monitoring for abuse of permissions by authorised system users
 - Monitoring security network access,
 - Analysing security system logs, security tools, and data
 - Review security alerts and taking steps to protect the information,
- **WA0204 Respond and recover**
 - Plan for disaster recovery and create contingency plans in the event of any security breaches
 - Respond to security events
 - Fix detected vulnerabilities to maintain a high-security standard.

- Apply technical knowledge in data loss prevention
- Remediating security issues

Supporting Evidence

- SE0201 Risk assessment and logs.
- SE0202 Report on findings in suitable format

3.3 Contextualised Workplace Knowledge

1. Organisational policies, procedures and guidelines which relate to cybersecurity.
2. Standard templates, documentation and tools.
3. Business processes and workflows pertaining to cybersecurity.
4. Team members.
5. Company specific operating systems.
6. Vendor specific cybersecurity tools and platforms.
7. Company specific best practices, quality standards and security protocols.
8. Version control system.
9. Sources of information available for testing.
10. Approval process

3.4 Criteria for Workplace Approval

Physical Requirements:

- Tools, equipment, systems, e.g.: company systems, documents, data, relevant meetings, teams and supervisors, design studio, etc.
- Key processes, e.g.: Cybersecurity design, testing and deployment processes project on the go or completed.

Human Resource Requirements:

- Maximum mentor/learner ratio of 1:3 in the ideal situation
- Supervisor/mentor: 2 years' software development experience

Legal Requirements:

- Legal (product) licences to use software
- OHS compliance certificate
- Ethical clearance (where necessary)

3.5 Additional Assignments to be Assessed Externally

None

SOURCES CONSULTED:

<https://aisn.net/5-cybersecurity-incident-response-steps/>

<https://www.simplilearn.com/cyber-security/ceh-certification>

<https://www.betterteam.com/information-security-analyst-job-description>

<https://www.thebalancecareers.com/information-security-analyst-skills-2062409>

<https://www.nist.gov/cyberframework/online-learning/five-functions>

SECTION 4: STATEMENT OF WORK EXPERIENCE

Curriculum Number:	252901001
Curriculum Title:	Cybersecurity Analyst

Learner Details	
Name:	
ID Number:	

Employer Details	
Company Name:	
Address:	
Supervisor Name:	
Work Telephone:	
E-Mail:	

252901001-WM-01, Compliance with Legal Cybersecurity Requirements, NQF Level 5, Credits 12

WM-01-WE01	Attend induction program and familiarise self with company processes, procedures, tools and culture		
	Scope of Work Experience	Date	Signature
WA0101	Attend induction program and familiarise self with the culture of the company.		
WA0102	Apply protocols and work etiquette.		
WA0103	Attend company specific information sharing sessions (e.g. standing meetings, toolbox talks, power hours, etc.)		
WA0104	Familiarise self with and apply “working from anywhere” protocols.		
WA0105	Read and understand company cybersecurity policy, protocols and procedures.		
WA0106	Comply with governance protocols and code of ethics of the company and ensure legal compliance by adhering to legal requirements (incl. but not limited to privacy, confidentiality, security of data, etc.).		
WA0107	Spend time in the various departments of the company, observe process flows and compile wire diagrams or workflow of the processes observed using suitable tools and showing the relationships and influences each of the departments have on each other.		
WA0108	Understand management requirements and expectations from cybersecurity measures.		
WA0109	Understand cybersecurity protocols and procedures.		
WA0110	Understand company assets in terms of cybersecurity.		
WA0111	Manage timesheets and apply self-management skills.		
WA0112	Collaborate with team members to achieve common and individual goals		
	Supporting Evidence	Date	Signature
SE0101	Attendance registers		

SE0102	Wire diagrams or workflows		
WM-01-WE02	Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.		
	Scope of Work Experience	Date	Signature
WA0201	Comply with requirements of risk management frameworks		
WA0202	Comply with security regulations and standards		
WA0203	Monitor compliance with information security policies and procedures		
WA0204	Give advice and guidance to staff on issues such as spam and unwanted or malicious emails.		
WA0205	Train organisation on security measures		
WA0206	Maintain an information security risk register and assist with internal and external audits relating to information security		
WA0207	Administering information security software and controls		
WA0208	Maintaining security records of monitoring and incident response activities		
WA0209	Document security breaches and assess the damage they cause.		
WA0210	Producing situational and incident-related reports		
WA0211	Providing timely and relevant security reports		
WA0212	Generate reports for both technical and non-technical staff and stakeholders		
	Supporting Evidence	Date	Signature
SE0201	Logs and technical documentation		
SE0202	Reports		
WM-01-WE03	Conduct the following tasks under supervision.		
	Scope of Work Experience	Date	Signature

WA0301	Comply with requirements of risk management frameworks		
WA0302	Comply with security regulations and standards		
WA0303	Monitor compliance with information security policies and procedures		
WA0304	Give advice and guidance to staff on issues such as spam and unwanted or malicious emails.		
WA0305	Train organisation on security measures		
WA0306	Maintain an information security risk register and assist with internal and external audits relating to information security		
WA0307	Administering information security software and controls		
WA0308	Maintaining security records of monitoring and incident response activities		
WA0309	Document security breaches and assess the damage they cause.		
WA0310	Producing situational and incident-related reports		
WA0311	Providing timely and relevant security reports		
WA0312	Generate reports for both technical and non-technical staff and stakeholders		
	Supporting Evidence	Date	Signature
SE0301	Logs and technical documentation		
SE0302	Reports		

	Contextualised Workplace Knowledge	Date	Signature
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	Additional Assignments to be Assessed Externally	Date	Signature
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252901001-WM-02, Cybersecurity Risk Assessment and Mitigation, NQF Level 5, Credits 20

WM-02-WE01	Shadow and observe an experienced Cybersecurity Analyst undertaking a cybersecurity risk assessment.		
	Scope of Work Experience	Date	Signature
WA0101	Determine information value		
WA0102	Identify and prioritise assets		
WA0103	Identify cyber threats		
	Performing cyber and technical threat analyses		
	Research/evaluate emerging cyber security threats and ways to manage them		
	Evaluate and deconstruct malware software		
	Stay current on it security trends and news.		
	Stay up-to-date on information technology trends and security standards.		
WA0104	Identify vulnerabilities		
	Discovering vulnerabilities in information systems		
	Identify potential weaknesses and implement measures, such as firewalls and encryption		
	Performing tests and risk analysis		
WA0105	Analyse controls and the need for new controls		
	Stay one step ahead of cyber attacks		
WA0106	Calculate the likelihood and impact of various scenarios on a per-year basis		
WA0107	Prioritize risks based on the cost of prevention vs information value		
WA0108	Document results in risk assessment report		

	Supporting Evidence	Date	Signature
SE0101	Attendance registers		
WM-02-WE02	Conduct a cybersecurity risk assessment under supervision.		
	Scope of Work Experience	Date	Signature
WA0201	Determine information value		
WA0202	Identify and prioritise assets		
WA0203	Identify cyber threats		
	Performing cyber and technical threat analyses		
	Research/evaluate emerging cyber security threats and ways to manage them		
	Evaluate and deconstruct malware software		
	Stay current on it security trends and news.		
	Stay up-to-date on information technology trends and security standards.		
WA0204	Identify vulnerabilities		
	Discovering vulnerabilities in information systems		
	Identify potential weaknesses and implement measures, such as firewalls and encryption		
	Performing tests and risk analysis		
WA0205	Analyse controls and the need for new controls		
	Stay one step ahead of cyber attacks		
WA0206	Calculate the likelihood and impact of various scenarios on a per-year basis		
WA0207	Prioritize risks based on the cost of prevention vs information value		
WA0208	Document results in risk assessment report		

	Supporting Evidence	Date	Signature
SE0201	Attendance register		
SE0201	Report on findings in suitable format		

	Contextualised Workplace Knowledge	Date	Signature

	Additional Assignments to be Assessed Externally	Date	Signature

252901001-WM-03, Cybersecurity Detection, Protection and Prevention Processes, NQF Level 5, Credits 20

WM-03-WE01	Shadow and observe an experienced Cybersecurity Analyst undertaking the following tasks.		
	Scope of Work Experience	Date	Signature
WA0101	Protect		
	Implement intrusion detection systems (IDS)		
	Implement security information event management (SIEM)		
	Implement threat plans and protocols,		
	Perform penetration testing		
	Perform security monitoring		
	Upgrade existing security systems		
	Engage in 'ethical hacking', for example, simulating security breaches		

WA0102	Prevent		
	Apply intrusion prevention tools		
	Help colleagues install security software and understand information security management.		
	Implement and maintain security frameworks for existing and new systems		
	Install, update and maintain security measures and operate software to protect systems and information infrastructure, including firewalls and data encryption programs.		
	Improve security efficiency		
WA0103	Detect		
	Use analytic tools to determine emerging threat patterns and vulnerabilities		
	Investigate security alerts and provide incident response		
	Investigate security breaches and other cybersecurity incidents.		
	Monitor and respond to 'phishing' emails and 'pharming' activity		
	Monitor computer networks for security issues.		
	Monitor for attacks, intrusions and unusual, unauthorised or illegal activity		
	Monitor identity and access management, including monitoring for abuse of permissions by authorised system users		
	Monitoring security network access,		
	Analysing security system logs, security tools, and data		
	Review security alerts and taking steps to protect the information,		
WA0104	Respond and recover		
	Plan for disaster recovery and create contingency plans in the event of any security breaches		

	Respond to security events		
	Fix detected vulnerabilities to maintain a high-security standard.		
	Apply technical knowledge in data loss prevention		
	Supporting Evidence	Date	Signature
SE0101	Attendance registers		
WM-03-WE02	Conduct a cybersecurity risk assessment under supervision.		
	Scope of Work Experience	Date	Signature
WA0201	Protect		
	Implement intrusion detection systems (IDS)		
	Implement security information event management (SIEM)		
	Implement threat plans and protocols,		
	Perform penetration testing		
	Perform security monitoring		
	Upgrade existing security systems		
	Engage in 'ethical hacking', for example, simulating security breaches		
WA0202	Prevent		
	Apply intrusion prevention tools		
	Help colleagues install security software and understand information security management.		
	Implement and maintain security frameworks for existing and new systems		
	Install, update and maintain security measures and operate software to protect systems and information infrastructure, including firewalls and data encryption programs.		

	Improve security efficiency		
	Prevent hacker intrusion		
WA0203	Detect		
	Use analytic tools to determine emerging threat patterns and vulnerabilities		
	Investigate security alerts and provide incident response		
	Investigate security breaches and other cybersecurity incidents.		
	Monitor and respond to 'phishing' emails and 'pharming' activity		
	Monitor computer networks for security issues.		
	Monitor for attacks, intrusions and unusual, unauthorised or illegal activity		
	Monitor identity and access management, including monitoring for abuse of permissions by authorised system users		
	Monitoring security network access,		
	Analysing security system logs, security tools, and data		
	Review security alerts and taking steps to protect the information,		
WA0204	Respond and recover		
	Plan for disaster recovery and create contingency plans in the event of any security breaches		
	Respond to security events		
	Fix detected vulnerabilities to maintain a high-security standard.		
	Apply technical knowledge in data loss prevention		
	Remediating security issues		
	Supporting Evidence	Date	Signature

SE0201	Attendance register		
SE0202	Report on findings in suitable format		

	Contextualised Workplace Knowledge	Date	Signature
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	Additional Assignments to be Assessed Externally	Date	Signature
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